

Evaluating Corporate Debts with Complex Liability Structures and Debt Covenants under the Jump-Diffusion Process

Abstract

A default-free bond can be evaluated independently from other outstanding bonds of the same issuer as their value has no effects on it. However, a risky bond issued by a defaultable issuer is different as the existence of other outstanding bonds of the same issuer will impact its value. Furthermore, the complex liability structures and bond covenants significantly influence the prices of risky bonds. This increases the complexity of pricing and in turn makes analytical formulas intractable. The popular structural model is a credit model that models the evolution of the asset value and specifies the conditions leading to default. This paper prices risky bonds by incorporating the complex liability structure and bond covenants into the structural model. An efficient tree model is then provided to price risky bonds. Because of its accuracy, this numerical model is of great help to exploring how credit spreads are influenced by the debt covenants and the change of the liability structure due to bond repayments or bond issuances. The jump-diffusion process is known to fit the empirical data better for its ability to generate large credit spreads for short-term corporate bonds and rich dynamics of the term structure of credit spreads. It can also be incorporated into our tree model.

Keywords: credit risk, structural model, tree, nonlinearity error, jump-diffusion process.

1 Introduction

Pricing a default-free bond can be done simply by summing all the future cash flows discounted by the appropriate risk-free rates. We do not need to consider the existence of other outstanding bonds issued by the same issuer as they do not affect default-free bonds. A risky bond, however, cannot be evaluated independently from other outstanding bonds of the same issuer because they may change the financial status of the issuer and the likelihood of default (see Lando, 2004).

Models are required to price risky bonds. There are two kinds of models: the reduced model and the structural model. This paper will focus on the structural model, which is first proposed by Black and Scholes (1973). It models the evolution of the firm's total asset value and specifies the conditions leading to default. Merton (1974) assumes that the firm's asset value follows a lognormal diffusion process and the default can only occur at the debt's maturity when the firm's asset value cannot meet this single debt. Thus the claims of equity holders can be viewed as call options on firm's asset and priced by the Black-Scholes formula accordingly (see Black and Scholes, 1973). To model the default prior to maturity, Black and Cox (1976) develop the first-passage model, which assumes that the firm defaults once its value falls below a liability-dependent default boundary. For example, the liability-dependent default boundary can be a constant proportion of the debt's

face value (as in Brennan and Schwartz, 1978; Nielsen et al., 2001; Kim et al., 1993; Longstaff and Schwartz, 1995) or the discounted present value of the debt's cash flow (as in Black and Cox, 1976; Briys and De Varenne, 1997). However, most of the literature focuses on the single-liability case, and it is hard to extend their results to price risky bonds for more complex liability structures.

Common bond covenants include restrictions on asset sales, default boundaries, seniorities of bonds, and early redemption (like putable and callable bonds). The values of risky bonds strongly depend on the assumptions regarding asset sales (see Lando, 2004). To protect the bond holder, bond covenants may prohibit the equity holders from selling the firm's assets to finance the coupon, debt, or dividend payouts; as a result, the equity holders have to finance the payments by issuing new equities. We call it the no-asset-sale assumption. This assumption incurs little mathematical difficulty as the evolution of the firm's asset value does not change with the coupon, debt, or dividend payouts. Thus it is widely used to keep the problem tractable analytically (see Leland, 1994). But allowing asset sales is more common in the real world. To keep the problem still tractable analytically, most academic papers adopt the proportional-asset-sale assumption, which allows the firm to sell a predetermined proportion of the firm's asset value (see Leland, 1994; Kim et al., 1993; Hilberink and Rogers, 2002). Under the proportional-asset-sale assumption, if the predetermined proportion of firm's asset value is more than the payout, the extra payment goes to the equity holders as dividends. On the other hand, if the predetermined proportion of firm's asset value cannot meet the payout, the equity holders will finance the shortfall by selling additional equities. Of course, the coupon or debt repayment is seldom set equal to a predetermined proportion to the firm's asset value contractually. Besides the two aforementioned assumptions, Merton (1974) and Brennan and Schwartz (1978) assume that the payout can be fully financed by selling the firm's asset, and we call it the total-asset-sale assumption. This assumption significantly increases the analytical complexity since the firm's asset value now jumps due to the repayment of bonds (see Lando, 2004). One may argue that the equity value under the total-asset-sale assumption can be priced by the approximate analytical formulas for pricing stock options with dividend payout (see Roll, 1977; Geske and Shastri, 1985; Heath and Jarrow, 1988), where the firm's asset value and the payout by selling the firm's asset are mapped to the stock price and dividend payout, respectively. However, these approximations could lead to unreasonable options prices, especially for barrier options according to Frishling (2002). The numerical results in our paper also confirm the claim.

The positive net-worth covenant forces the firm to default if its asset value falls below a certain threshold called the exogenous default boundary. This covenant can prevent the equity holders from increasing the equity value at the expense of the debt value by investing the firm's asset on risky projects (see Leland, 1994). Since the default boundary is liability-dependent, a complex liability structure implies a complex default boundary, which makes the evaluation intractable. For example, assume a defaultable firm issues two bonds with maturity dates t_1 and t_2 , where $t_1 < t_2$. Following Brennan and Schwartz (1978), the default boundary prior to t_1 should be the sum of the principal values of the two bonds, and the exogenous default boundary between t_1 and t_2 should be the principal value of the bond maturing at t_2 . If there is no positive net-worth covenant in the bond indenture, the default boundaries are determined endogenously based on the assumptions on asset sales. Under the no-asset-sale and proportional-asset-sale assumptions, the firm defaults if the equity holders fail to raise enough equity capital to meet the firm's debt obligations (see Leland, 1994). But under the total-asset-sale assumption, the equity holders will repay the bonds or coupons as long as there are sufficient assets to cover the payments. This is because even if this leaves the firm's asset

value lower, the equity holders at least retain an option with a positive value, whereas default leaves them with nothing.

Seniority refers to the order of repayments in the event of bankruptcy. When the issuer goes bankrupt, a senior bond must be repaid before a subordinated bond; therefore, a senior bond has lower risk and a higher price than an otherwise subordinated debt. However, the numerical results in our paper suggest that the liability structure of the firm may influence the value of the bonds. Consider the case that a subordinated bond and a senior one with maturity dates t_1 and t_2 , respectively, where $t_1 < t_2$. Under the total-asset-sale assumption, the repayment of the subordinated bond at t_1 may significantly influence the financial status of the firm and the riskiness of the senior bond.

Pricing a risky bond in the presence of other outstanding bonds seems to be first studied by Geske (1977). Under the no-asset-sale assumption, he argues that the equity holders would repay the short-term bond at its maturity date t_1 if the equity value at t_1 is larger than the repayment of the short-term bond. Thus the equity value and the long-term bond can be priced as a compound option if there is no positive net-worth covenant. However, pricing can be difficult under different bond covenants. For example, under the total-asset-sale assumption, pricing a risky bond with a positive net-worth covenant and complex bond repayment schedules is equivalent to pricing a time-varying barrier option on a stock with dividends.

A callable bond allows the issuer to redeem the bond at a specified call price prior to maturity, whereas a puttable bond allows the bond holder to redeem the bond at a specified put price prior to maturity. For a puttable bond, the bond holder will sell back the bond to the issuer if it is more beneficial for the bond holder to exercise the right than keep the bond. Similarly, an issuer will call the bond if it is more beneficial to do so. The aforementioned callable and puttable strategies are hard to evaluate analytically with complex liability structures and bond covenants in place, but they can be easily implemented by our tree model.

The first major contribution of this paper is to provide a flexible tree model for pricing risky bonds with complex liability structures and debt covenants. To model the jumps of the firm's asset value because of the coupon or debt repayment, we adopt the trinomial structure (see Dai, 2009). To eliminate the price oscillations for barrier options for tree models (see Figlewski and Gao, 1999), we follow Ritchken (1995) to make certain nodes or price levels on our tree model align with the varying default boundaries. The early redemption and seniority of bonds can be easily implemented by the standard backward induction. The added flexibility of the tree allows the firm's asset value to be discretely monitored rather than continuously monitored, which reflects the fact that the debt holders can only monitor the firm's asset value at the releases of financial reports or significant announcements (see Duffie and Lando, 2001).

This paper also incorporates the jump-diffusion process into our tree model. Zhou (2001) and Hilberink and Rogers (2002) add stochastic jump events to the firm value process. The jumps may be caused by incomplete market information, lawsuits, or sudden financial turmoil. This addition can generate large credit spreads for short-term corporate bonds and various shapes of credit spread curves that fit the real world data better (see Zhou, 2001). The jump-diffusion feature can be easily implemented by merging the trinomial structure mentioned in Dai (2009) into the Hilliard and Schwartz's bivariate tree (see Hilliard and Schwartz, 2005). Numerical results suggest that our tree not only generates large credit spreads on short-term corporate bonds and many shapes of credit spreads, but also gives reasonable bond prices that reflect different complex liability structure and debt covenants.

Our paper is organized as follows. The mathematical model, the debt covenants, the structure of tree models, and the oscillation problem are introduced in Section 2. Section 3 describes how various bond covenants and the jump-diffusion process are implemented by our tree models. Backward induction is described in Section 4. Two concrete examples are given to demonstrate the tree construction and backward induction in Section 5. Section 6 analyzes the prices of risky bonds for complex liability structures and different bond covenants. Section 7 concludes the paper.

2 Basic Terms and Preliminaries

2.1 The Dynamics of the Firm's Asset Value

Denote the firm's asset value at time t as V_t . We follow Zhou (2001) by assuming that the firm's asset value follows a jump-diffusion process as follows:

$$dV_t = ((r - \lambda \bar{k}) V_t - P) dt + \sigma V_t dz + k V_t dq, \quad (1)$$

where r is the risk-free rate, P denotes the payout from selling firm's asset continuously to finance debt and dividend payments, σ denotes the volatility contributed by the smooth diffusion component, and dz denotes the Wiener process. The jump event is governed by the compound Poisson process, where k denotes the magnitude of random jumps, and q denotes a Poisson process with an intensity λ . The process of the firm's asset value can be assumed to follow a lognormal process by setting $\lambda = 0$. Note that the distribution of jump magnitude k follows $\ln(1+k) \sim N(\gamma, \delta^2)$, and its mean $\bar{k} \equiv E(k) = e^{\gamma+0.5\delta^2} - 1$. The standard Brownian motion, the random jump magnitude, and the Poisson process are assumed to be independent. Thus the total volatility of firm's asset value σ_V^2 (contributed by both diffusion component and jump component) satisfies

$$\sigma_V^2 = \sigma^2 + \lambda E k^2 = \sigma^2 + \lambda(e^{2\delta^2} - 2e^{0.5\delta^2} + 1) \approx \sigma^2 + \lambda\delta^2. \quad (2)$$

The process of the firm's asset value can be influenced by the assumptions on asset sales. We first assume that the tax shelter benefit is not considered for simplicity. Under the no-asset-sale assumption, P is set to 0; that is, the debt repayment or dividend payout are financed by selling additional equities and would not influence the process of firm's asset value. Allowing on asset sales would increase the mathematical difficulty to model the process of the firm's asset value. If the payout is made continuously and the payout per unit of time is C , then $P \equiv C$ under the total-asset-sale assumption (see Merton, 1974; Brennan and Schwartz, 1978). Under the proportional-asset-sale assumption, a predetermined proportion D of asset value is sold to finance the payout (i.e., $P \equiv DV_t$), where the proportion D can be assumed to depend on the payout (e.g., $D \equiv \frac{C}{V_0}$ (see Leland, 1994)). On the other hand, if the payout C is made discretely at time t , then we have $V_t = V_{t-} - C$ under the total-asset-sale assumption and $V_t = V_{t-} - V_{t-}D$ under the proportional-asset-sale assumption. If the tax shelter benefit is considered, then the tax benefit realized by paying interest expenses should be added back to the firm's asset value.

The default event is triggered once the firm's asset value falls below a certain threshold, called the default boundary. For a firm that issues bonds with positive net-worth covenants, the boundary at time t are exogenously specified as a function of firm's outstanding bonds at time t , like a constant proportion of the sum of outstanding bonds' face values (see Nielsen et al., 2001; Kim et al., 1993;

Longstaff and Schwartz, 1995), or the discounted present value of outstanding bonds (see Black and Cox, 1976; Briys and De Varenne, 1997). The default boundary can also be endogenously defined. For example, the default event is triggered once the payout DV_t cannot meet current debt obligation in Kim et al. (1993), and it is triggered if the firm fails to raise sufficient equity capital to meet current debt obligation in Leland (1994).

2.2 Background Knowledge for Tree Construction and the Oscillation Problem

Constructing a Lattice under the Lognormal-Diffusion Process

A tree is a popular numerical model for modeling the evolution of a stochastic process. It partitions the time span from time 0 to time T into n equal time steps and specifies the value of the stochastic process at each time step. Thus the length of one time step Δt equals T/n . Each node on a tree can branch into ℓ nodes at the next time step, which is called an ℓ -nomial tree. For example, the nodes and black branches in Fig. 1(a) forms a 2-time-step binomial tree (i.e., $\ell = 2$). At each time step, the firm value V can either make an up move to become Vu with probability P_u or a down move to become Vd with probability $P_d \equiv 1 - P_u$.

The firm's asset value in most literature is assumed to follow the lognormal diffusion process. For convenience, define the V -log-price of the firm's asset value V' as $\ln(V'/V)$ and the log-distance between firm's asset values V and V' as $|\ln(V) - \ln(V')|$. The jump-diffusion process can be degenerated into the lognormal diffusion process by substituting $\lambda = 0$ in Eq. (1). The mean (denoted as μ) and variance (denoted as Var) of the V_t -log-price of $V_{t+\Delta t}$ are $(r - 0.5\sigma^2) \Delta t$,¹ and $\sigma^2 \Delta t$, respectively. To make the tree converges to the process of the firm's asset value as $n \rightarrow \infty$, μ and Var are matched by adjusting the branching probabilities (like P_u and P_d) and multiplication factors (like u and d). For example, the CRR tree (see Cox et al., 1979) adopts this solution: $u = e^{\sigma\sqrt{\Delta t}}$, $d = e^{-\sigma\sqrt{\Delta t}}$, $P_u = \frac{e^{(\mu+0.5\sigma^2)\Delta t}-d}{u-d}$ and $P_d = \frac{e^{(\mu+0.5\sigma^2)\Delta t}-u}{d-u}$. Thus the log-distance between any two adjacent nodes (e.g., nodes X and Y in Fig. 1(a)) is $2\sigma\sqrt{\Delta t}$. To deal with the jump of firm's asset value due to the debt repayment or dividend payout, we use the trinomial structure developed in Dai (2009). This trinomial structure can be used to connect an arbitrary node to the nodes at the next time step if the nodes at the next time step follow the CRR structure. Take node W with asset value Vu in Fig. 1(a) for example. Selling firm's asset P would reduce the asset value from Vu (node W) to $Vu - P$ (node Q). Assume that among all the nodes at time T , the $(Vu - P)$ -log-price of node Y is the nearest to μ . Then the branches from node Q will connect to node Y and its two adjacent nodes: X and Z . To make the trinomial branches calibrate μ and Var , Dai (2009) proves that the feasible solution for the branching probabilities (p_u , p_m , and p_d) can be uniquely solved by the Crammer's rule. Our paper will show that their proof can be further extended to make the trinomial structure more flexible for tree construction.

The Bivariate Lattice for the Jump-Diffusion Process

Hilliard and Schwartz (2005) decompose the V_t -log-price of $V_{t+\Delta t}$ into the diffusion component and

¹Here we focus on the no-asset-sale assumption. The total-asset-sale and proportional-asset-sale assumption are modeled by the trinomial structure that will be introduced later.

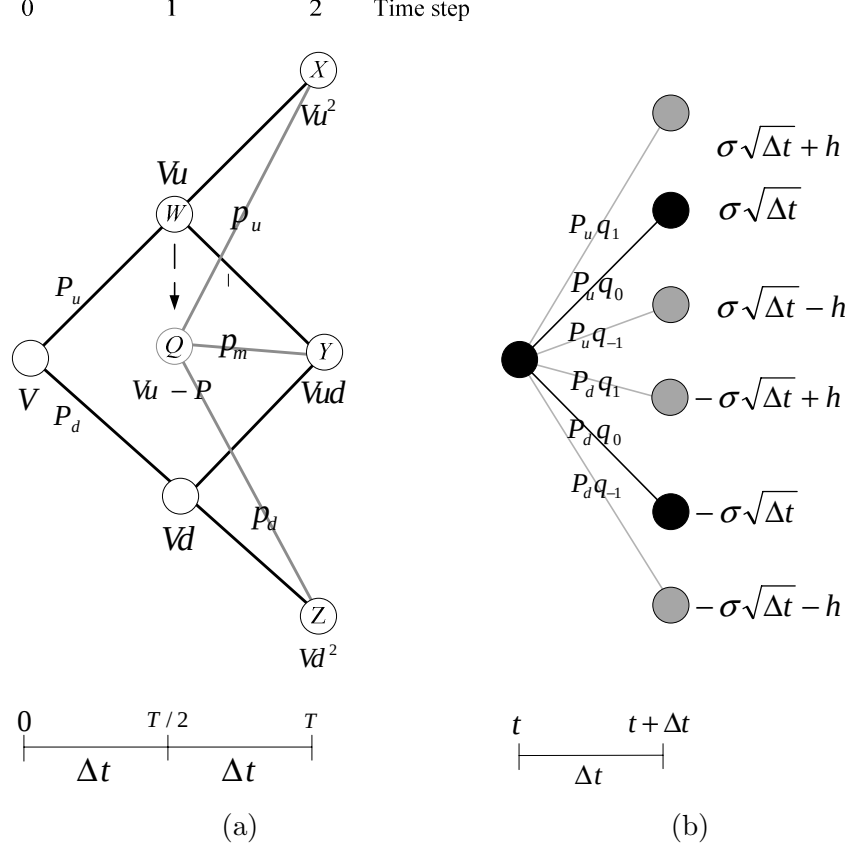


Figure 1: **Tree Structures.** Panel (a) describes the binomial tree and the trinomial branches for modeling the lognormal diffusion process. The nodes and black branches denote the structure of the binomial tree. The gray branches denote the trinomial structure discussed in Dai (2009). P_u and P_d denote the branching probabilities of the binomial structure, and p_u , p_m , and p_d denote the branching probabilities of the trinomial structure. The firm's asset value at each node is listed next to the node. Panel (b) illustrates the structure of a bivariate tree for modeling the jump-diffusion process. The V_t -log-price for each node at time $t + \Delta t$ is listed next to the node.

jump component by rewriting Eq. (1) as follows:

$$M_{\Delta t} \equiv \ln \left(\frac{V_{t+\Delta t}}{V_t} \right) \equiv [(r - \lambda \bar{k} - 0.5\sigma^2) \Delta t + \sigma z(\Delta t)] + \left[\sum_{i=0}^{n(\Delta t)} \ln(1 + k_i) \right], \quad (3)$$

where $n(\Delta t)$ denotes the number of jump occurs during the time interval $[t, t + \Delta t]$, and k_i denotes the size of the i -th jump.² In their bivariate tree, the diffusion component is modeled by the aforementioned CRR tree except that μ is replaced by $(r - \lambda \bar{k} - 0.5\sigma^2) \Delta t$.

The jump component is modeled by positioning extra $4m$ nodes to match the first $2m$ local moments of the lognormal jumps. Motivated by the decomposition in Eq. (3), the random variable $M_{\Delta t}$ can be discretely approximated as follows: $V_{i\Delta t}$ -log-prices of the $2 \times (2m + 1)$ nodes at time step $i + 1$ following node $M_{i\Delta t}$ can be represented by

$$M_{\Delta t} \approx c\sigma\sqrt{\Delta t} + jh, \quad c = \pm 1, j = 0, \pm 1, \pm 2, \dots, \pm m. \quad (4)$$

²Here we assume $P = 0$ for simplicity.

Above, $c \in \{-1, 1\}$ denotes the upward or the downward movement of the firm value driven by the diffusion component (called the diffusion move), and j denotes the number of positions above or below $c\sigma\sqrt{\Delta t}$ driven by the jump component (called the jump move). Note that $j = 0$ corresponds to the diffusion moves. For example, a 1-time-step bivariate tree with $m = 1$ is illustrated in Fig. 1(b), where the black nodes and branches denote the diffusion moves, and the gray nodes and branches denote the jump moves. The magnitude of the basic jump unit is set to

$$h = \sqrt{\gamma^2 + \delta^2}. \quad (5)$$

The probabilities for the jump component q_j ($j = 0, \pm 1, \pm 2, \dots, \pm m$) are chosen to match the first $2m$ local moments of the jump component as follows:

$$\sum_{j=-m}^{j=m} (jh)^i q_j = \mathbb{E} \left[\sum_{w=0}^{n(\Delta t)} \ln(1 + k_w) \right]^i, \quad i = 1, 2, \dots, 2m. \quad (6)$$

In addition,

$$\sum_{j=-m}^{j=m} q_j = 1$$

must hold. By solving the above $2m + 1$ equations, we can obtain $2m + 1$ probabilities q_j ($j = 0, \pm 1, \pm 2, \dots, \pm m$). Since the diffusion component and the jump component are assumed to be independent, the probability for $M_{\Delta t} = \sigma\sqrt{\Delta t} + jh$ is $P_u q_j$ and the probability for $M_{\Delta t} = -\sigma\sqrt{\Delta t} + jh$ is $P_d q_j$.

The Nonlinearity Error

Theoretically, the pricing result generated by the tree converges to the theoretical value of a contingent claim (like equity or bond value) as $n \rightarrow \infty$ (Duffie, 1996); however, the pricing result may oscillate significantly. Figlewski and Gao (1999) argue that this phenomenon is mainly due to the nonlinearity error — the error introduced by the nonlinearity of the value function of the contingent claim. The nonlinearity error can be significantly reduced by making a node or a price level of the tree coincide with the locations (called critical locations) where the value function of the contingent claim is highly nonlinear. For the structural model, critical locations occur along the default boundary and time points when debt (or dividend) payout occurs. Our tree model can adapt its structure to make a node, a time step, or a price level of the tree to coincide with the critical locations; thus it can significantly suppress the nonlinearity error.

3 Tree Construction

We first explain and extend the proof for the trinomial structure introduced in Dai (2009). Next, we describe how to use this trinomial structure to construct a tree to model complex liability structures, assumptions on asset sales, and positive net-worth covenants.³ Finally, we show how to incorporate the jump-diffusion process into our tree model. This section focuses on the total-asset-sale assumption

³Other bond covenants, like seniority and early redemption, are discussed in Section 4.

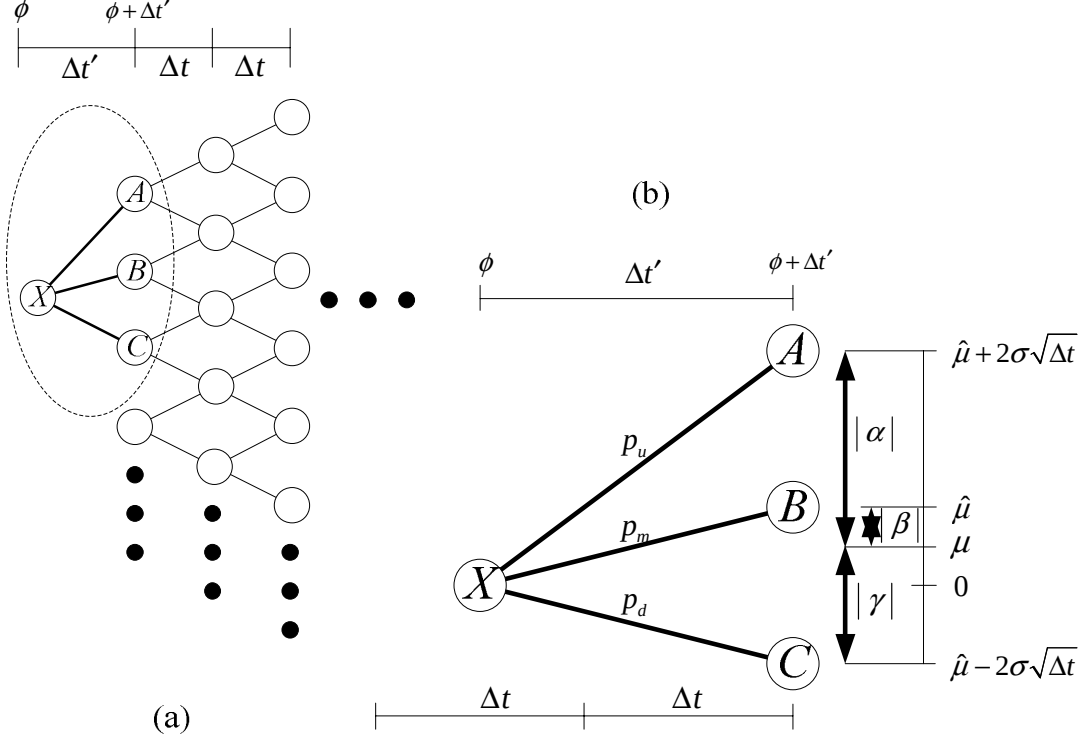


Figure 2: The Trinomial Structure. Panel (a) illustrates that there exists a unique trinomial structure for an arbitrary node X at time ϕ to connect to 3 following nodes A , B , and C at time $\phi + \Delta t'$. The tree structure enclosed by an ellipse is zoomed in in Panel (b). In Panel (b), the firm's asset value can move from node X to nodes A , B , and C with probability p_u , p_m , and p_d , respectively. The $v(X)$ -log-price of each node and the log-distance between the $v(X)$ -log-price of each node and μ (see Eq. (7)) (i.e., $|\alpha|$, $|\beta|$, and $|\gamma|$) are illustrated in the rightmost part of Panel (b). Above, μ denotes the mean of the $v(X)$ -log-price of $V_{t+\Delta t'}$ (see Eq. (7)). The $v(X)$ -log-prices of $v(X)$, $v(B)$, $v(A)$, and $v(C)$ are 0, $\hat{\mu}$, $\hat{\mu} + 2\sigma\sqrt{\Delta t}$, and $\hat{\mu} - 2\sigma\sqrt{\Delta t}$, respectively. The length of the time step for the trinomial structure, $\Delta t'$, satisfies the constraint $\Delta t \leq \Delta t' < 2\Delta t$.

since it significantly increases the analytical complexity to model the firm's asset value (see Lando, 2004). The extension to the other two assumptions on asset sales are straightforward.

3.1 The Trinomial Structure

In Fig. 2(a), we assume that the nodes at time $\phi + \Delta t'$ begins a truncated CRR tree with the length of each time step being equal to Δt . Dai (2009) argues that there always exists a feasible trinomial structure for the arbitrage node X at time ϕ that connects to 3 following nodes (nodes A , B , and C in the figure) at time $\phi + \Delta t'$ under the condition $\Delta t' = \Delta t$. Our paper relaxes this constraint to $\Delta t \leq \Delta t' < 2\Delta t$; therefore, our tree model can provide more flexibility to fit critical locations (discussed later). Denote the firm's asset value of node Z as $v(Z)$ for convenience. Note that the trinomial branches should match the mean μ and variance Var of the $v(X)$ -log-price of $V_{\phi+\Delta t'}$, where

$$\mu \equiv (r - \lambda \bar{k} - 0.5\sigma^2) \Delta t', \quad (7)$$

$$\text{Var} \equiv \sigma^2 \Delta t'. \quad (8)$$

And the branching probabilities p_u , p_m , and p_d (see Fig. 2(b)) are between 0 and 1.

Note that the log-distance between any two adjacent nodes at time $\phi + \Delta t'$ is $2\sigma\sqrt{\Delta t}$. Thus, at time $\phi + \Delta t'$, there must exist a unique node B whose $v(X)$ -log-price $\hat{\mu}$ lies in the interval $[\mu - \sigma\sqrt{\Delta t}, \mu + \sigma\sqrt{\Delta t}]$. We select node B and its two adjacent nodes A and C to construct a valid trinomial structure from node X . Define α , β , and γ as the differences between μ and the $v(X)$ -log-price of $v(A)$, $v(B)$, and $v(C)$, respectively:

$$\begin{aligned}\beta &\equiv \hat{\mu} - \mu, \\ \alpha &\equiv \hat{\mu} + 2\sigma\sqrt{\Delta t} - \mu = \beta + 2\sigma\sqrt{\Delta t}, \\ \gamma &\equiv \hat{\mu} - 2\sigma\sqrt{\Delta t} - \mu = \beta - 2\sigma\sqrt{\Delta t}.\end{aligned}$$

The branching probabilities from node X (i.e., p_u , p_m , p_d) can be derived by solving the following 3 equalities:

$$p_u\alpha + p_m\beta + p_d\gamma = 0, \quad (9)$$

$$p_u(\alpha)^2 + p_m(\beta)^2 + p_d(\gamma)^2 = \text{Var}, \quad (10)$$

$$p_u + p_m + p_d = 1, \quad (11)$$

Equations (9) and (10) match the mean and variance of the $v(X)$ -log-price of $V_{t+\Delta t'}$ and Eq. (11) ensures that the probabilities p_u , p_m , p_d do sum to one. Appendix A shows that the above 3 equations indeed yield valid branching probabilities under the relaxed assumption $\Delta t \leq \Delta t' < 2\Delta t$.

3.2 Tree Construction: a Typical Example

A typical case with the total-asset-sale assumption and constant exogenous default boundaries is considered here to demonstrate how our tree is developed by combining the trinomial structure with the CRR tree. We focus on the case that the two zero-coupon bonds with face values F_1 and F_2 are repaid discretely at time $0.5T$ and T (see Fig. 3). Under the total-asset-sale assumption, the bond repayments are fully financed by selling the firm's asset – Lando (2004) argues that this significantly increases the difficulty to model the process of the firm's asset value. Section 3.3 will discuss how our tree can be extended to other assumptions on asset sales. We assume that a positive net-worth covenant is enforced, and the firm is forced to default once its asset value is lower than a constant proportion κ of the face value of outstanding bonds.⁴ Thus the default boundary for the time interval $[0, 0.5T]$ is $\kappa(F_1 + F_2)$ (the thick black line in Fig. 3), and the default boundary for the time interval $[0.5T, T]$ is κF_2 (the thick gray line). To suppress the oscillation problem, our tree makes two price levels coincide with these two default boundaries. In addition, the firm defaults when it fails to meet the debt obligation F_2 at time T . Thus there should be a node (node F) to coincide with the value F_2 at time T . Section 3.3 will show that our tree can be extended to other exogenous default boundaries. If there is no positive net-worth covenant in the bond indenture, the default event is triggered when the equity holders fail to raise enough equity capital to meet the debt obligation. The default boundary will be determined endogenously by the backward induction mentioned in Section 4.

The tree in Fig. 3 can be decomposed into two parts: the structure in the time interval $[0, 0.5T]$

⁴The default boundaries in Nielsen et al. (2001); Kim et al. (1993); Longstaff and Schwartz (1995) follow this setting.

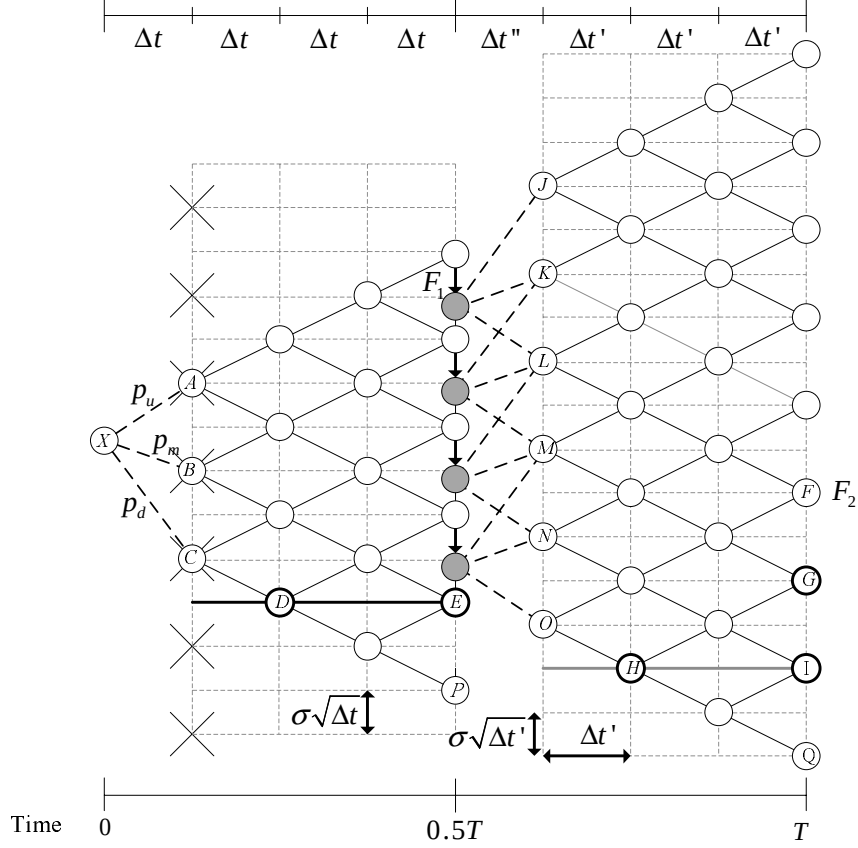


Figure 3: Modeling the Firm's Asset Value with Positive Net-Worth Covenants under the Total-Asset-Sale Assumption. The tree models the firm's asset value that is allowed to sell its asset to finance bond repayments. Assume the firm issues two zero-coupon bonds: one matures at time $0.5T$ with face value F_1 and the other one matures at time T with face value F_2 . The firm's initial asset value at time 0 is represented by node X . The black and gray thick lines denote the exogenous default boundaries due to positive net-worth covenants. The firm defaults once its value reaches nodes marked by the thick black circles. At time $0.5T$, the nodes above the default boundary jump downward to the corresponding gray nodes to reflect that the firm sells its asset to finance the bond repayment F_1 . To reduce the nonlinearity error, the tree should have two price levels to coincide with the default boundaries and a node (F in our example) to coincide with the debt obligation F_2 at time T . To make one price level coincide with the black thick line, the nodes connected by the node X must be some of those nodes marked by the crosses at time Δt . To make another price level coincide with the gray thick line and a node match the debt obligation F_2 at time T simultaneously, the length of a time step (i.e., $\Delta t'$ and $\Delta t''$) is adjusted. The dashed lines represent the trinomial structures mentioned in Section 3.1. The truncated CRR trees from time Δt to $0.5T$ and from time $0.5T + \Delta t''$ to T are laid on the left-hand side and right-hand side grids, respectively. Nodes A , B , and C are selected for constructing the trinomial structure. p_u , p_m , and p_d denote the branching probabilities from node X .

and the structure in $[0.5T, T]$. We focus on the tree construction of the former part first. The firm's initial asset value at time 0 is represented by node X . The tree rooted at node X must have a price level that coincides with the default boundary (marked by the thick black line). Assume that the time interval $[0, 0.5T]$ is divided into n_1 time steps, there exists a n_1 -time-step tree that fits the aforementioned requirements as illustrated in Fig. 3, where the length of each time step $\Delta t = \frac{0.5T}{n_1}$. For example, $n_1 = 4$ and $\Delta t = T/8$ in Fig. 3.⁵ Then the CRR tree with first two time steps being truncated in time interval $[\Delta t, 0.5T]$ is laid on the grid that partitions the plane into cells. The cell width of the grid is Δt , and the cell height of the grid is $\sigma\sqrt{\Delta t}$ (both measured in the log-distance). To make a price level of the CRR tree coincide with the default boundary $\kappa(F_1 + F_2)$, we lay out the grid from the default boundary upward. Thus all the nodes that are connected by the branches of node X at time Δt must be some of the intersections of the grid marked by the crosses. Denote the $v(X)$ -log-price of the default boundary as $\ell(\equiv \ln\left(\frac{\kappa(F_1+F_2)}{v(X)}\right))$ for convenience. Then the $v(X)$ -log-price of all possible candidate nodes at time Δt (marked by the crosses) are represented by

$$\begin{cases} \ell + 2j\sigma\sqrt{\Delta t}, & \text{if the truncated CRR tree has an even number of time steps,} \\ \ell + (2j+1)\sigma\sqrt{\Delta t}, & \text{otherwise,} \end{cases}$$

for some integer j . For example, the truncated CRR tree in Fig. 3 has 3 time steps, and the $v(X)$ -log-price of the candidate nodes at time Δt can be represented by $\ell + (2j+1)\sigma\sqrt{\Delta t}$. The trinomial branches from node X can be constructed by the procedure in Section 3.1. Assume that the $v(X)$ -log-price of node B is within the range $[\mu - \sigma\sqrt{\Delta t}, \mu + \sigma\sqrt{\Delta t})$. Then the trinomial branches from node X will connect node B and its two adjacent nodes A and C . The probabilities for the trinomial branches (i.e., p_u , p_m , and p_d) can be solved by the formulas in Appendix A. The truncated CRR tree grows from node A , B , and C at time Δt to the nodes at time $0.5T$. The default boundary passes through nodes D and E , and the firm defaults once the firm's asset value falls below the default boundary.

At time $0.5T$, the down jumps from the white nodes to the gray ones with amount F_1 denote that the firm sells its asset to repay the debt obligation. The evolution of the firm's asset value for the latter part of the tree begins from these gray nodes. These gray nodes will connect to a truncated CRR tree beginning at time $0.5T + \Delta t''$ by the trinomial branches in order to avoid the combinatorial explosion problem.⁶ This truncated CRR tree is again laid on the grid that partitions the plane into cells. The width and height of each cell are $\Delta t'$ and $\sigma\sqrt{\Delta t'}$, respectively. To suppress the nonlinearity error, this part of the tree must simultaneously hit the default boundary (marked by the thick gray line) and the debt obligation F_2 (node F) at time T . This implies the log-distance between F_2 and κF_2 (i.e., the default boundary) must be integral multiples of $\sigma\sqrt{\Delta t'}$; this can be achieved by picking a proper $\Delta t'$. Define the $v(X)$ -log price of F_2 and κF_2 as h' and ℓ' , respectively. Then we pick $\Delta t' = (\frac{h'-\ell'}{2\eta\sigma})^2$, where $\eta \equiv \left\lceil \frac{h'-\ell'}{2\sigma\sqrt{\Delta t}} \right\rceil$. For example, $\eta = 2$ in Fig. 3. Note that the time interval $[0.5T, T]$ might not be divided into integral time steps if the length of a time step is $\Delta t'$. To solve this problem, we assume that the truncated CRR tree has $\lfloor \frac{0.5T}{\Delta t'} \rfloor - 1$ time steps. Thus the length of the first time step for the latter part of the tree, $\Delta t''$, is the remaining amount of time to

⁵In our simple example, $n_1 = 4$ in Fig. 3.

⁶Dai (2009) argues that the number of nodes of the tree will grow drastically if the trees following the gray nodes are not recombined.

make the whole time span (of the latter part of the tree) $0.5T$, i.e.,

$$\Delta t'' = 0.5T - \left(\left\lfloor \frac{0.5T}{\Delta t'} \right\rfloor - 1 \right) \Delta t'.$$

Clearly, $\Delta t' \leq \Delta t'' < 2\Delta t'$, so the trinomial structure introduced in Section 3.1 can be used to connect the gray nodes at time $0.5T$ to the nodes at time $0.5T + \Delta t''$ by the same node selection procedure used in constructing the trinomial structure for node X . The truncated CRR tree grows from nodes J, K, L, M, N , and O to the nodes at time T . The firm defaults once the firm's asset value falls below the default boundary or F_2 at time T .

The effect of the tax benefit and bankruptcy cost (see Leland, 1994) can be incorporated into our tree without difficulty. To incorporate the tax benefit into our tree model, the jump of firm's asset value at time $0.5T$ would be F_1 minus the tax benefit received from repaying the short-term bond. The bankruptcy cost is incurred when the firm defaults. It is subtracted from the firm's asset value and the remaining value is paid to the bond holders. The discrete default boundary (due to regular releases of financial reports or the information asymmetry mentioned in Duffie and Lando (2001)) can be easily implemented by our tree model. If the firm's asset value is only discretely monitored at bond repayment dates, then the firm defaults once its asset value reach nodes E, P, G, I , and Q .

3.3 Extensions to Other Bond Covenants

Our tree construction methodology can be easily extended to deal with different assumptions regarding asset sales and exogenous default boundaries as illustrated in Fig. 4. The stair tree proposed in Dai (2009) can be used to model different assumptions regarding asset sales as illustrated in Panel (a). The firm's asset value at node X at time step 0 will move to either node A or B at time step 1. P_1 and P_2 denote the payouts financed by selling the firm's asset. Under the total-asset-sale assumption with a continuous payout C per year (i.e., $P \equiv C$ in Eq. (1))⁷, both P_1 and P_2 are set to $C\Delta t$. Under the proportional-asset-sale assumption by selling a fixed proportion D of firm's asset value (i.e., $P = DV_t$ in Eq. (1)), $P_1 \equiv DV_u\Delta t$ and $P_2 \equiv DV_d\Delta t$. Under the no-asset-sale assumption, both P_1 and P_2 are set to 0. The branches from node a follow the CRR binomial branches. All the white nodes at time step 2 follow the structure of the CRR tree, so the branches from node b and the branching probabilities p_u, p_m , and p_d can be constructed by the trinomial structure construction mentioned in Section 3.1.

Panel (b) models a time-varying default boundary marked by the thick black curve. To make the curve hit by the nodes on the tree, we make a price level (marked by the crosses) at each time step coincide with the default boundary. All other nodes are laid out from the gray crosses upward and the log-distance between any two adjacent crosses at the same time step is $2\sigma\sqrt{\Delta t}$. Let node X denote the initial firm's asset value at time 0. Thus the successor nodes connected by node X can be selected from the crosses at time step 1 by the node selection strategy mentioned in Section 3.1. Again, the successor nodes for these 3 nodes can be selected from the crosses at time step 2 by the aforementioned procedure. The above procedure can be repeatedly applied to complete the whole structure of the tree. Note that the firm defaults once its asset value reaches the gray nodes (i.e., the default boundary); therefore, there are no successor nodes for these gray nodes.

⁷This assumption can be found in Merton (1974); Brennan and Schwartz (1978).

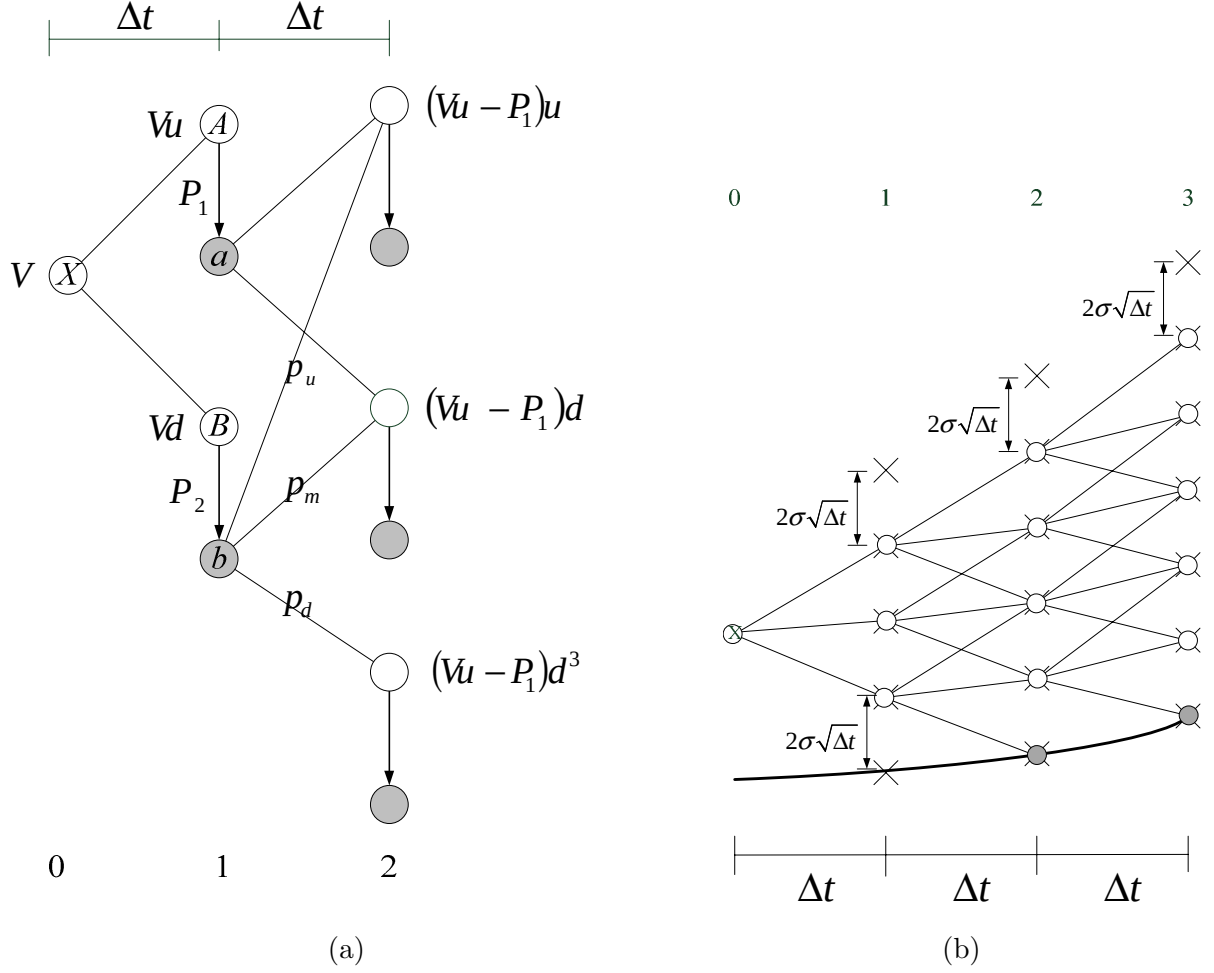


Figure 4: **Extensions to Other Bond Covenants.** In Panel (a), the white nodes denote the firm's asset value before selling the asset to finance the debt (or dividend) payout. The gray nodes denote the asset value after selling the asset. The value for each white node is listed next to the node. P_1 and P_2 denote the payouts financed by selling the firm's asset at nodes A and B , respectively. p_u , p_m , and p_d denote the probabilities of the branches from node b . Panel (b) models a time-varying default boundary marked by the thick black curve. The log-distance between any two adjacent crosses at the same time step is $2\sigma\sqrt{\Delta t}$. All the tree nodes must be located on the crosses. The firm defaults when its asset value reaches the gray nodes.

3.4 Extension to the Jump-Diffusion Process

If the firm's asset value follows the jump-diffusion process, a tree for pricing risky bonds with positive net-worth covenants can be constructed by incorporating the trinomial structure into the modified version of the bivariate tree mentioned in Hilliard and Schwartz (2005) (see Fig. 5(b)). We follow Eq. (3) by decomposing each time step into the diffusion phase (for modeling the diffusion component in Eq. (3)) and the jump phase (for modeling the jump component). This decomposition is feasible since the diffusion process (driven by the Brownian motion), the jump magnitudes, and the occurrences of the jump events (driven by the Poisson process) in Eq. (1) are independent.

The tree can be constructed by the following two stages. In the first stage, we construct a tree for modeling the diffusion component of the firm's asset value. By following the tree construction procedure mentioned in Section 3.2, we construct a tree that a price level on it coincides with the default boundary (marked by the thick gray line) as illustrated in Fig. 5(a). The tree nodes (colored in white) involved in the first stage are called diffusion nodes for simplicity. In the second stage, we add the jump phase to each time step as illustrated in Fig. 5(b). The nodes (colored in gray) called jump nodes are added to reflect the jump events. For example, the flat movement from node A to node a with probability q_0 reflects that no jump event occurs during the first time step. The movement from node A to node a_1 with probability q_1 reflects that an upward jump occurs during the first time step, where the log-distance between node a_1 and node a is h (see Eq. (5)). Similarly, the movement to node a_{-1} with probability q_{-1} denotes a downward jump occurs during the first time step. For simplicity, the parameter m in Eq. (4) is set to one in our example. Thus there are two jump nodes following node A and the branching probabilities q_1 , q_0 , and q_{-1} calibrate the first two moments of the jump component (see Eq. (6)). Hilliard and Schwartz (2005) argue that the accuracy for modeling the jump component can be significantly improved by increasing m mildly, and our tree can take a larger m without difficulty. The branches from the jump nodes at time step 1 model the diffusion component during the diffusion phase of the second time step. The trinomial structures developed in Section 3.1 are adopted to connect these jump nodes to the diffusion nodes at the next time step. Note that additional diffusion nodes D and I are added to meet the connection requirements for node a_1 and c_{-1} . The firm defaults when its asset value falls below node h .

4 Backward Induction Procedure

The equity value (E), bond value (B), tax benefit (TB), and bankruptcy cost (BC) can be viewed as contingent claims on firm's asset value and computed on the tree model by backward induction. Many bond covenants, like the endogenous default boundary (due to the inability to raise enough equity capital to meet the debt obligation), seniority of bonds, and early redemption can all be implemented by designing sophisticated backward induction procedures. In this section, we focus a typical case that the firm issues two bonds: B_1 matures at time T_1 with face value F_1 and B_2 matures at time T_2 with F_2 , where $T_1 < T_2$. The bond B_1 (or B_2) is assumed to pay coupon C_1 (or C_2) continuously. Thus the coupon payout for B_1 (or B_2) over a time step with length Δt is $C_1\Delta t$ (or $C_2\Delta t$). The backward induction for an arbitrary node at time t can be divided into the following four cases: (1) $t = T_2$, (2) $T_1 < t < T_2$, (3) $t = T_1$, and (4) $0 \leq t < T_1$. If the firm issues more than two bonds, the backward induction can be divided into more than four cases according to the maturity dates of these bonds. For convenience, denote $E(t, V)$, $B_1(t, V)$, and $B_2(t, V)$ as the values

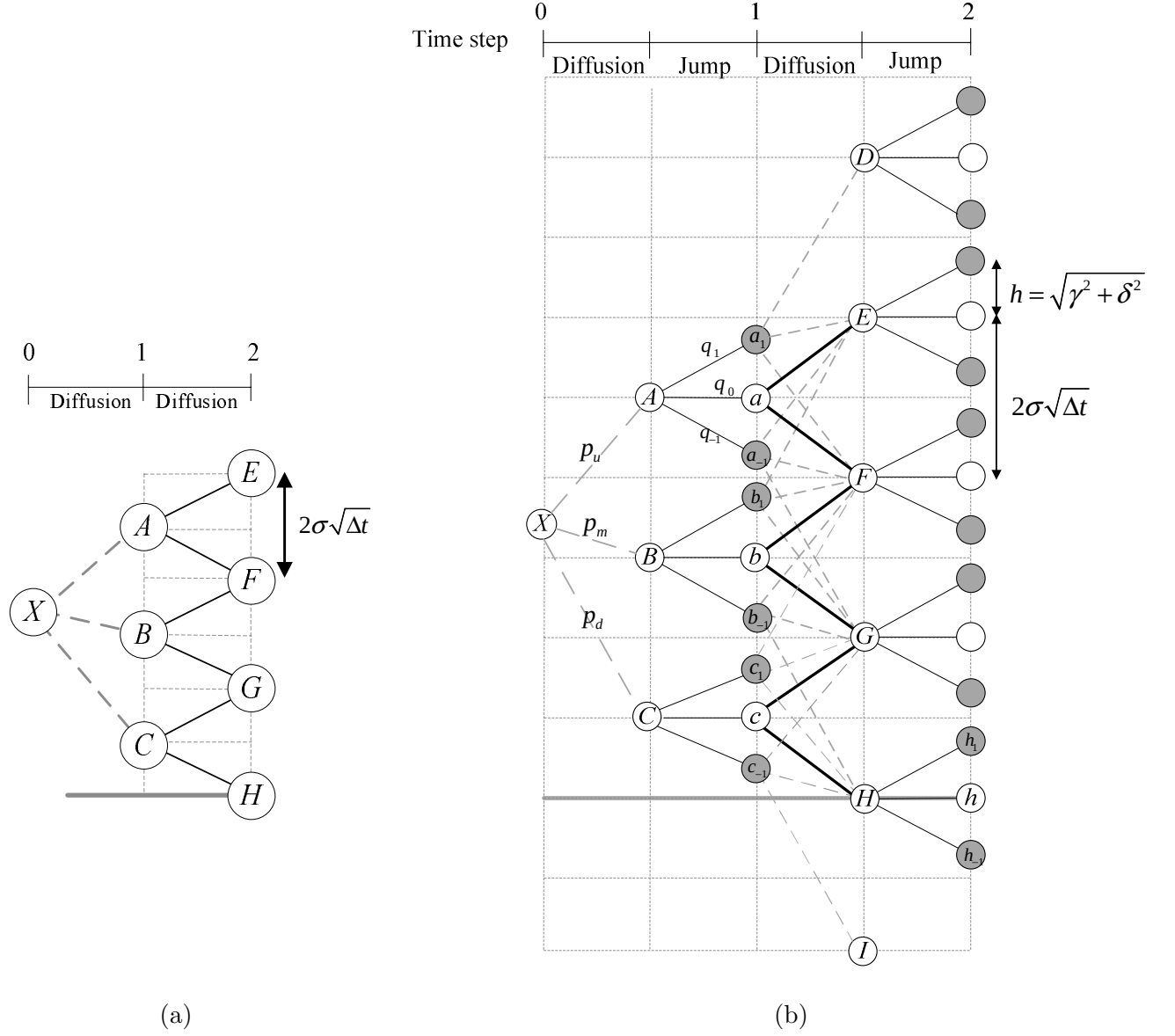


Figure 5: Pricing Risky Bonds with an Exogenous Default Boundary under the Jump-Diffusion Process. Panel (a) illustrates the first stage of the tree construction: modeling the diffusion component. Panel (b) illustrates the second stage of tree construction: adding the jump component. The thick gray solid lines denote the default boundary. In Panel (b), each time step is divided into the diffusion phase and the jump phase. The white nodes and gray nodes represent the nodes for the diffusion component (called diffusion nodes) and for the jump component (called jump nodes), respectively. The solid black lines denote the CRR binomial structure, and the dashed gray lines denote the trinomial structure mentioned in Section 3.1. All diffusion nodes except node X lie on the intersections of the grid marked by the dotted lines and the log-distance between two adjacent diffusion nodes at the same time step is $2\sigma\sqrt{\Delta t}$. The branches from diffusion nodes except node X follow the CRR binomial structure. The branches from jump nodes follow the trinomial structure. The branching probabilities for the jump nodes (i.e., q_1 , q_0 , and q_{-1}) are defined in Eq. (6). The magnitude of the basic jump unit h is defined in Eq. (5).

of equity, bond B_1 , and bond B_2 at time t given the firm's asset value V .

Case 1: $t = T_2$

At B_2 's maturity date T_2 , the default occurs when the firm's asset value V cannot meet the B_2 's face value F_2 plus the coupon payment over the last time step $C_2\Delta t$. Thus we have

$$E(T_2, V) = \begin{cases} V - F_2 - (1 - \tau)C_2\Delta t & \text{if } V \geq F_2 + (1 - \tau)C_2\Delta t \\ 0 & \text{otherwise,} \end{cases}, \quad (12)$$

and

$$B_2(T_2, V) = \begin{cases} F_2 + C_2\Delta t & \text{if } V \geq F_2 + (1 - \tau)C_2\Delta t \\ (1 - \alpha)V & \text{otherwise,} \end{cases},$$

where $(1 - \alpha)$ denotes the recovery rate (i.e., α is the bankruptcy cost) and τ denotes the tax rate. $\tau C_2\Delta t$ in Eq. (12) denotes the tax shelter benefit from repaying the coupon. Note that TB and BC are equal to 0 and αV , respectively, when the firm defaults at time T_2 ; otherwise, TB and BC are equal to τC_2 and 0, respectively. Note that the permanent capital structure (i.e., infinite maturity debt) can be approximated by a finite-maturity model with a very large T .

Case 2: $T_1 < t < T_2$

The backward induction is required to obtain the value of the contingent claims at time t by the values of the contingent claims at the next time step (i.e., at time $t + \Delta t$). The firm's asset value at each time step should be adjusted due to selling the firm's asset for financing the debt or dividend payout at that time step. We use the superscripts “-” and “+” to denote the times and the firm's asset values before and after the payout, respectively. Note that the occurrence of the adjustment implies that the firm does not default at that time step. The value of any contingent claim on the after-adjustment asset value V^+ at time t^+ , denoted as $f(t^+, V^+)$, can be expressed as the discounted expected value of that contingent claim at the next time step. Take node b at time t in Fig. 4(a) for example, we have the discounted expected value:

$$\begin{aligned} f(t^+, Vd - P_2) &\equiv e^{-r\Delta t} (p_u \times f((t + \Delta t)^-, (Vu - P_1)u) + p_m \times f((t + \Delta t)^-, (Vu - P_1)d) \\ &\quad + p_d \times f((t + \Delta t)^-, (Vu - P_1)d^3)). \end{aligned}$$

Whether the debt payout is carried out or not depends on the firm's status (survive or default) at time t . Therefore, the value of a contingent claim before the payout (i.e., at time t^-) can be expressed in terms of the value of the contingent claim after payout (i.e., at time t^+) as follows:

$$E(t^-, V^-) = \begin{cases} E(t^+, V^+) - S & \text{if default does not occur.} \\ 0 & \text{if default occurs.} \end{cases}, \quad (13)$$

and

$$B_2(t^-, V^-) = \begin{cases} B_2(t^+, V^+) + C_2\Delta t & \text{if default does not occur.} \\ (1 - \alpha)V^- & \text{if default occurs.} \end{cases},$$

where S denotes the payout financed by selling additional equities. S and the evolution of the firm's asset value depend on the assumptions on asset sales. Under the no-asset-sale assumption, the equity holders sell additional equity to finance after-tax coupon (i.e., $S = (1 - \tau)C_2\Delta t$) and the firm's asset value does not change (i.e., $V^+ = V^-$). Under the proportional-asset-sale assumption, the firm is

allowed to sell a predetermined proportion D of firm's asset to finance the coupon and dividend payouts, and then $S = (1 - \tau)C_2\Delta t - DV^-\Delta t$ and $V^+ = V^- - DV^-\Delta t$. Note that a negative S implies that the payout by selling firm's asset exceeds the coupon payment, and the extra payout goes to the equity holders as dividends. Under the total-asset-sale assumption, the firm is allowed to sell asset to fully finance the payouts. Thus we have $S = 0$ and $V^+ = V^- - (1 - \tau)C_2\Delta t$.

The default boundary can be either exogenously or endogenously defined. If a positive net-worth covenant exists, the default boundary is triggered exogenously when the firm's asset value falls below a certain exogenous default boundary V_B (i.e., $V^- < V_B$). Otherwise, the default event can be triggered endogenously by the inability of the equity holders to raise sufficient equity capital to meet the payout (i.e., $E(t^+, V^+) < S$).

Case 3: $t = T_1$

The firm is required to repay the face value and the coupon of B_1 at time T_1 . The equity value $E(t^-, V^-)$ can be evaluated by following Eq. (13) except that the term S and the evolution of the firm value are slightly changed. Under the no-asset-sale assumption, the equity holders sell additional equity to finance after-tax coupons of B_1 and B_2 and the face value of B_1 (i.e., $S = (1 - \tau)(C_1 + C_2)\Delta t + F_1$) and the firm's asset value does not change (i.e., $V^+ = V^-$). Under the proportional-asset-sale assumption, $S = (1 - \tau)(C_1 + C_2)\Delta t + F_1 - DV^-\Delta t$ and $V^+ = V^- - DV^-\Delta t$. Under the total-asset-sale assumption, $S = 0$ and $V^+ = V^- - (1 - \tau)(C_1 + C_2)\Delta t - F_1$. The default conditions are similar to those discussed in Case 2. For example, under the no-asset-sale assumption, if there is no positive net-worth covenant in the bond indenture, the firm defaults when $E(t^+, V^+) < S = (1 - \tau)(C_1 + C_2)\Delta t + F_1$.

The seniorities of B_1 and B_2 will influence the payoffs of them when default occurs. Let $B_i \succ B_j$ denote that B_i is more senior than B_j . Then we have

$$B_2(t^-, V^-) = \begin{cases} B_2(t^+, V^+) + C_2\Delta t & \text{if default does not occur.} \\ \max(\min((1 - \alpha)V^- - (F_1 + C_1\Delta t), PV(t^-, B_2)), 0) & \text{if default occurs and } B_1 \succ B_2 \\ \min((1 - \alpha)V^-, PV(t^-, B_2)) & \text{if default occurs and } B_2 \succ B_1 \end{cases}, \quad (14)$$

and

$$B_1(t^-, V^-) = \begin{cases} F_1 + C_1\Delta t & \text{if default does not occur.} \\ \min((1 - \alpha)V^-, F_1 + C_1\Delta t) & \text{if default occurs and } B_1 \succ B_2 \\ \max(\min((1 - \alpha)V^- - PV(t^-, B_2), PV(t^-, B_1)), 0) & \text{if default occurs and } B_2 \succ B_1 \end{cases},$$

where $PV(t, B_i)$ denotes the present value of all future unpaid payments of bond B_i at time t .

Case 4: $0 \leq t < T_1$

$E(t^-, V^-)$ can be evaluated by following Eq. (13) except that the term S and the evolution of the firm value are slightly changed. Under the no-asset-sale assumption, $S = (1 - \tau)(C_1 + C_2)\Delta t$ and $V^+ = V^-$. Under the proportional-asset-sale assumption, $S = (1 - \tau)(C_1 + C_2)\Delta t - DV^-\Delta t$ and $V^+ = V^- - DV^-\Delta t$. Under the total-asset-sale assumption, $S = 0$ and $V^+ = V^- - (1 - \tau)(C_1 + C_2)\Delta t$. The default conditions are similar to those in Case 2. $B_2(t^-, V^-)$ and $B_1(t^-, V^-)$ can be evaluated

by the following formulas:

$$B_2(t^-, V^-) = \begin{cases} B_2(t^+, V^+) + C_2 \Delta t & \text{if default does not occur.} \\ \max(\min((1 - \alpha)V^- - PV(t^-, B_1), PV(t^-, B_2)), 0) & \text{if default occurs and } B_1 \succ B_2 \\ \min((1 - \alpha)V^-, PV(t^-, B_2)) & \text{if default occurs and } B_2 \succ B_1 \end{cases},$$

and

$$B_1(t^-, V^-) = \begin{cases} B_1(t^+, V^+) + C_1 \Delta t & \text{if default does not occur.} \\ \min((1 - \alpha)V^-, PV(t^-, B_1)) & \text{if default occurs and } B_1 \succ B_2 \\ \max(\min((1 - \alpha)V^- - PV(t^-, B_2), PV(t^-, B_1)), 0) & \text{if default occurs and } B_2 \succ B_1 \end{cases}.$$

The early redemption can also be easily implemented by the backward induction procedure. For example, a puttable bond B_i will be sold back to the issuer with a put price $\mathcal{P}$ at time t if $B_i(t^-, V^-) < \mathcal{P}$. Similar arguments can be applied to price callable bonds. Additionally, the case that the coupon is paid discretely can be implemented without difficulty.

5 The Concrete Examples

Several numerical examples are given to illustrate how to price a risky bond issued by a firm with a complex liability structure under different assumptions on asset sales, seniorities, positive net-worth covenants, and early redemption. We assume that the firm issues two zero-coupon bonds: B_1 and B_2 , where B_1 with face value $F_1 = \$500$ matures at year 0.5 and B_2 with $F_2 = \$2,500$ matures at year 1. The firm's asset value is assumed to follow the lognormal diffusion process (i.e., $\lambda = 0$ in Eq. (2)). We then assume the initial firm's asset value is \$5,000, the risk-free interest rate is $r = 2\%$, and the firm volatility is $\sigma_V = 40\%$. Both the bankruptcy cost α and the tax rate τ are assumed to be 0 for simplicity. We first focus on pricing B_2 under the total-asset-sale assumption with a positive net-worth covenant (i.e., an exogenous default boundary) and demonstrate how the seniority and puttable feature affect the price of B_2 . Then we extend our analysis to the no-asset-sale assumption. Two four-step trees are constructed in Fig. 6 and 7 for these two assumptions, where the length of each time step Δt is 0.25 year. With the aforementioned setting, the multiplicative factors for the CRR tree are $u \approx 1.221$ and $d \approx 0.819$. The binomial branching probabilities for the CRR tree are $P_u \approx 0.463$ and $P_d \approx 0.537$.

5.1 An Example under the Total-Asset-Sale Assumption

A typical tree for modeling the firm's asset value under the total-asset-sale assumption is illustrated in Fig. 6. Note that the number in the white cell of a node denotes the firm's asset value at that node and the numbers in the gray-scale cells denote the values of B_2 under different scenarios. For simplicity, we assume that a prior-to-maturity default boundary \$2,400⁸ is discretely located at year 0.5. Another default boundary at B_2 's maturity is the face value of B_2 , \$2,500. To alleviate the price oscillations, we make two nodes E and H coincide with the default boundaries marked by the thick black lines. We first follow the procedures suggested in Section 3.2 by laying a grid marked by the

⁸Here we follow Brennan and Schwartz (1978) by assuming that the default boundary is a constant fraction (80%) of the sum of the face values of outstanding debts (i.e., $F_1 + F_2$).

thin dotted lines to coincide with the boundary \$2,400 at year 0.5. The nodes of the truncated CRR tree is now laid on the grid. The branches from node A connect the truncated CRR tree by the trinomial structure. If the firm does not default at year 0.5 like nodes B , C , and D , part of the firm's asset is sold to finance the debt repayment of B_1 ; that is, the firm's asset value at these nodes jumps down by \$500. Again, a grid marked by the thin dotted lines is laid to coincide with the boundary \$2,500 at year 1. The branches from nodes B , C , D also connect the truncated CRR tree by the trinomial structures. The branching probabilities for these trinomial structures are listed in Table 1.

In the following discussion, we focus on 3 scenarios: (1) The first scenario is that $B_2 \succ B_1$, (2) the second scenario is that $B_2 \succ B_1$ and B_2 is a putable bond with put price $F_2 e^{-\kappa(T-t)}$, where $\kappa = 4\%$, and (3) the third scenario is that $B_1 \succ B_2$.

Scenario 1: $B_2 \succ B_1$

Note that the number in the light-gray cell of a node denotes the value of B_2 at that node. At B_2 's maturity, the firm defaults (node I) if its asset value cannot meet the debt obligation \$2,500, and the holder of B_2 receives the remaining firm's asset value $\$1,675.8(1 - \alpha)$ (recall that the bankruptcy cost α is set to 0). The firm survives if its asset value exceeds B_2 's face value and the debt holder receives \$2,500. The debt value at year 0.75 is obtained by backward induction with the binomial branching probabilities P_u and P_d . For example, the value of B_2 at node F is

$$e^{-0.02 \times 0.25} \times (0.463 \times 2500 + 0.537 \times 1675.8) \approx 2046.83.$$

Two different cases might occur at year 0.5: (1) If the firm's asset value before repaying B_1 , V^- , is less than or equal to the default boundary \$2,400, the holder of B_2 receives $\min((1 - \alpha)V^-, \text{PV}(0.5^-, B_2))$ (recall Eq. (14)). For example, the bond value at node E is \$2,400 ($= (1 - \alpha)V^-$). (2) If V^- exceeds the boundary \$2,400 (like nodes B , C , and D), the value of B_2 is calculated by backward induction with the trinomial branching probabilities listed in Table 1. For example, the bond value at node D is

$$e^{-0.02 \times 0.25} \times (0.1173 \times 2487.53 + 0.7498 \times 2487.53 + 0.1329 \times 2046.83) \approx 2416.84.$$

Comparing to node C with firm's asset value \$4,841.3 and bond value \$2,475.13, a lower firm's asset value at node D , i.e., \$3,080.38, entails a higher default probability and a lower bond price \$2,416.84. Similarly, the bond value for each node at year 0 and 0.25 can be computed by backward induction. The current value of B_2 at node A priced by this four-step tree is

$$e^{-0.02 \times 0.25} \times (0.318 \times 2462.78 + 0.662 \times 2431.61 + 0.020 \times 2395.79) \approx 2428.63.$$

Scenario 2: $B_2 \succ B_1$ and B_2 is a putable bond with a put price $\mathcal{P}$

Note that the values of B_2 at nodes A , D , E , G , J under Scenario 2 are different from those under Scenario 1, and the bond values at these nodes are put in the mid-gray cells. The bond values of other nodes under Scenario 2 can be referred to the values listed in the light-gray cells of those nodes. We assume that the put price $\mathcal{P}(t) = F_2 e^{-\kappa(T-t)}$, where $\kappa = 4\%$. Thus B_2 will be sold back to the issuer with a put price $\mathcal{P}(t)$ at time t if $B_2(t^-, V^-) < \mathcal{P}(t)$, like nodes D , G , and F . For example, the value at node D is $\max(2416.84, 2500e^{-0.04 \times 0.5}) = 2450.50 = \$2,450.50$. Comparing the bond values of both Scenario 1 and 2 at node D , this putable feature provides protection for B_2 against the damage from repaying the subordinated bond B_1 . Note that although $B_2(0.75^-, 2046.83) = \$2,046.83 < \$2,475.12$ at node F , selling B_2 back to the issuer causes the firm to default and the holder of B_2 receives the

	The Branching Probabilities						
	A	A'	B	C	D	B'	C' D'
Upper (P_u)	0.3179	0.2439	0.1889	0.1173	0.3694		
Middle (P_m)	0.6615	0.7106	0.7368	0.7498	0.6207		
Lower (P_d)	0.0205	0.0455	0.0742	0.1329	0.0099		

Table 1: **The Trinomial Branching Probabilities in Figs. 6 and 7.** The branching probabilities for node A , B , C , D , A' , B' , C' , and D' can be obtained by solving Eqs. (9)–(11).

firm’s asset value \$2,046.83. Generally speaking, the putable feature of B_2 increases its value from \$2,428.63 (the light-gray cell of A) to \$2,441.10 (the mid-gray cell of A).

Scenario 3: $B_1 \succ B_2$

The values of B_2 at nodes A , D , E , G , J are listed in the dark-gray cells, and the values for other nodes are put in the light-gray cells for simplicity. Since B_2 is more subordinated than B_1 , the holder of B_2 will get the residue of the firm’s asset value after B_1 is repaid as the firm defaults. For instance, the firm defaults at node E since its asset value hits the default boundary \$2,400 at year 0.5, and the holder of B_2 receives $\max(\min(2400 - 500, PV(0.5^-, B_2)), 0) = \$1,900$ (recall Eq. (14)). Generally speaking, the value of B_2 is reduced from \$2,428.63 to \$2,423.32 due to the change of the bond’s seniority.

5.2 An Example under the No-Asset-Sale Assumption

A tree for modeling the firm’s asset value under the no-asset-sale assumption is illustrated in Fig. 7. All the numerical settings and three scenarios are the same as those in Section 5.1. To alleviate the oscillation problem, this tree also hits the default boundaries marked by the thick black lines. The tree in Fig. 7 is similar to the tree in Fig. 6 except that the firm’s asset value does not jump down by \$500 at year 0.5. This is because the repayment of B_1 must be financed by selling additional equities under the no-asset-sale assumption. Note that the firm may default if the equity holders fail to raise enough equity capital to meet the debt obligation. We list the “after-debt-repayment” equity values (i.e., $E(t^+, V^+)$ in Eq. (13)) in the ellipses for the nodes located after year 0.5 (inclusive) since these values will be used to determine the solvency of the firm.⁹ The equity value can be calculated by backward induction if the firm’s asset value is larger than the default boundary. Otherwise, the firm defaults and the equity value is set to 0 (like node E'). The following discussions will first focus on the aforementioned three scenarios with an exogenous default boundary. We then show this tree can be easily extended to price B_2 as there is no positive net-worth covenant; that is, there is no exogenous default boundary, and the default event is totally endogenously triggered.

Scenario 1: $B_2 \succ B_1$

The values of B_2 are illustrated in the light-gray cells. At year 1 and 0.75, the value of B_2 at each node is the same as that in Fig. 6. At year 0.5, there are two cases that cause the firm to default: (1) The firm’s asset value is less than the default boundary \$2,400 (like node E'), and (2) the firm’s asset value is larger than the default boundary, but the equity holder fails to raise sufficient equity capital to meet the debt obligation. Case (2) occurs when the exogenously default boundary is too low. If the firm defaults at a node with asset value V , the holder of B_2 receives $\min((1 - \alpha)V, PV(0.5^-, B_2))$

⁹In this simple case, the firm never defaults prior to year 0.5. This is because no debt repayment occurs before year 0.5 and no default boundary is located before year 0.5.

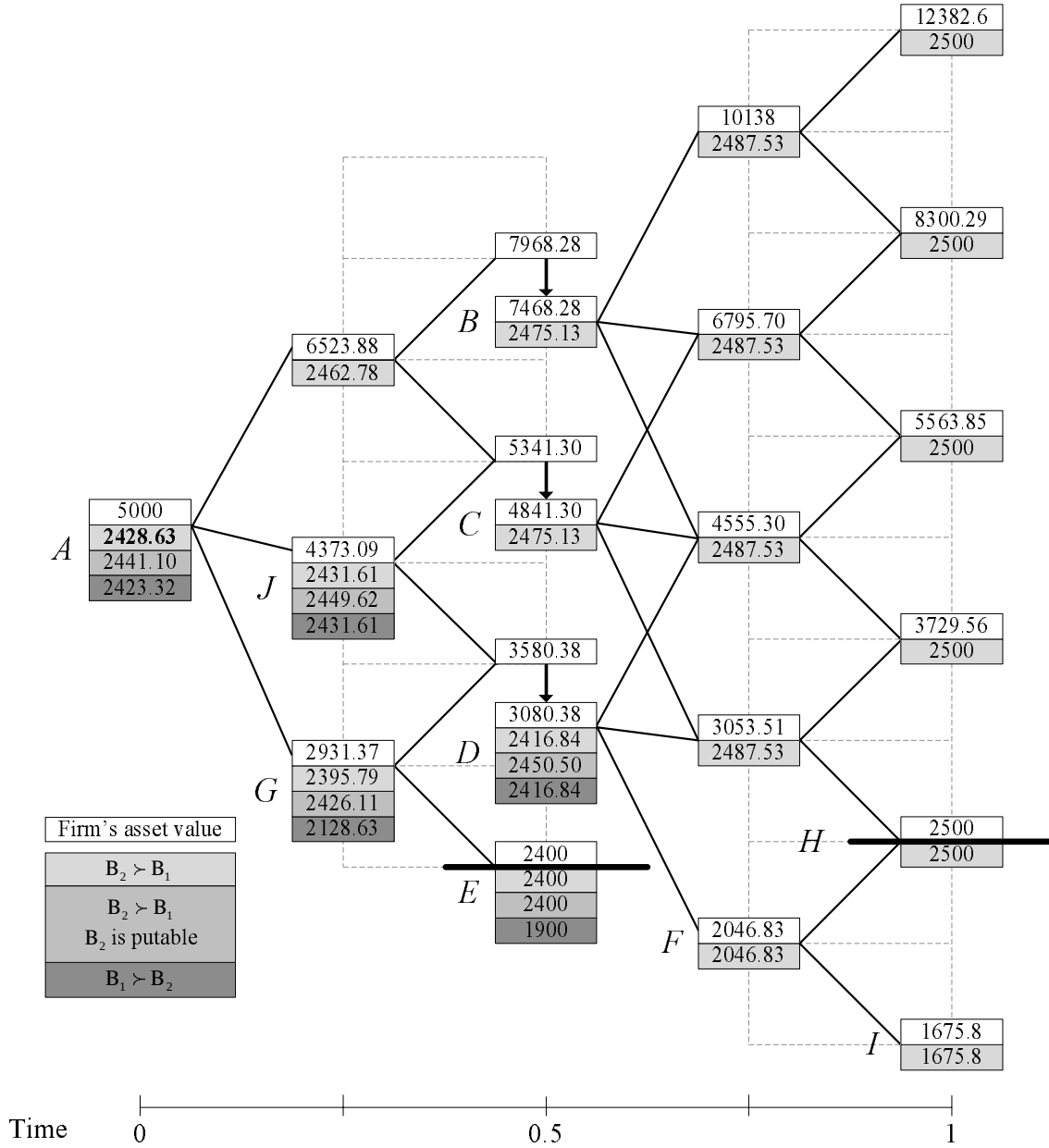


Figure 6: A Numerical Example under the Total-Asset-Sale Assumption. The firm issues two zero-coupon bonds: B_1 with face value $F_1 = \$500$ matures at year 0.5 and B_2 with face value $F_2 = \$2,500$ matures at year 1. The numbers in the white cells denote the firm's asset value. The thick black lines denote the default boundaries; the default boundary at year 0.5 is \$2,400 and that at year 1 is \$2,500. The numbers in the gray-scale cells denote the value of B_2 under different scenarios. The bonds values under Scenario 1 are listed in light-gray cells. The bonds values for Scenario 2 and Scenario 3 at nodes A , D , E , G , J are listed in mid-gray cells and dark-gray cells, respectively. The bonds values for these two scenarios at other nodes are listed in light-gray cells. The trinomial branching probabilities for nodes A , B , C , and D are listed in Table 1.

(recall Eq. (14)). On the other hand, if the firm does not default, the value of B_2 is calculated by backward induction with the trinomial branching probabilities listed in Table 1. For example, the firm survives at nodes B' , C' , and D' because the firm's asset values are larger than \$2,400 and the equity values after repaying F_1 at nodes B' , C' , and D' are all greater than \$500(= F_1) (i.e., $E(t^+, V^+) > F_1$). Note that the value of B_2 at node D \$2,416.84 (see Fig. 6) is less than that at node D' \$2,470.38 since repaying B_1 by selling the firm's asset does harm the value of B_2 . Finally, the value of B_2 at node A' \$2,448.13 is higher than that at node A \$2,428.63 since the no-sale-asset assumption provides better protection for the holder of B_2 than the total-asset-sale assumption does.

Scenario 2: $B_2 \succ B_1$ and B_2 is a puttable bond with a put price $\mathcal{P}$

The values of B_2 at nodes A' , G' and E' are listed in the mid-gray cells and the values for other nodes are put in the light-gray cells for simplicity. The calculation is similar to the calculation under the total-asset-assumption. B_2 will be sold back to the issuer with a put price $\mathcal{P}(t)$ at time t if $B_2(t^-, V^-) < \mathcal{P}(t)$ (like nodes G' , and F'). For example, the value at node G' is $\max(2420.64, 2500e^{-0.04 \times 0.75} = 2426.11) = \$2,426.11$. Note that the puttable feature of B_2 slightly increases its value from \$2,448.13 (the light-gray cell of node A') to \$2,448.24 (the mid-gray cell of A'). Comparing to the example in Fig. 6, the puttable feature can increase the value of B_2 more under the total-asset-sale assumption (from \$2,428.63 to \$2,441.10) because it can alleviate the damage on B_2 due to selling firm's asset to finance the repayment of B_1 . Specifically, B_2 's holder will sell the bond back to the issuer at node D in Fig. 6 to prevent the loss of bond value due to the down jump of the firm's asset value from \$3,580.38 to \$3,080.38. However, under the no-asset-sale assumption, B_2 's holder will keep the bond at node D' , the corresponding node of D in Fig. 7. Of course, both the restriction on asset sales and the puttable feature give the better protection of B_2 's holder, which can be confirmed by the highest bond value \$2448.24 than the bond values under all other scenarios.

Scenario 3: $B_1 \succ B_2$

The values of B_2 at nodes A' , G' and E' are listed in the dark-gray cells and the values for other nodes are put in the light gray cells. As B_1 is more senior than B_2 , B_2 is repaid after B_1 is repaid when the firm defaults. For example, the firm defaults at node E' and the calculation of B_2 's value at node E' is the same as the calculation for the value at node E in Fig. 6. The value of B_2 is reduced from \$2,448.13 to \$2,442.81 due to the change of the bond's seniority. Note that \$2,442.81 is higher than the bond value under the total-asset-sale assumption (i.e., \$2,423.32 in the dark-gray cell of node A) since financing the repayment of the senior bond B_1 by selling firm's asset harms the value of the subordinated bond B_2 .

This tree model can be easily extended to the case that there is no positive net-worth covenant in the bond indenture; that is, the exogenous default boundary \$2,400 at year 0.5 does no longer exist. Under this case, the "after-debt-repayment" equity value at node E' can be reevaluated to be \$238.19 by the Black-Scholes formula. Because the equity value \$238.19 is less than the repayment of B_1 \$500, the equity holders will not repay the loan and the firm defaults. Under this four-step tree, the firm defaults at nodes E' and I' in spite of the existence of the positive net-worth covenant. Therefore, even if there is no positive net-worth covenant, the value of B_2 is the same as that with a positive net-worth covenant in this tree model.

To sum up, our approach can price risky bonds under complex liability structures and various bond covenants. Table 2 tabulates the values of B_2 calculated by our tree with 1000 time steps. In the table, three cases under the no-asset-sale assumption are listed: (1) the exogenous default boundaries is assumed to be \$2,400, (2) the exogenous default is assumed to be \$3,000, and (3) the

	Scenario 1	Scenario 2	Scenario 3
Total-Asset-Sale	2405.97	2430.76	2402.43
No-Asset-Sale (1)	2435.47	2441.34	2425.42
No-Asset-Sale (2)	2438.07	2441.85	2427.07
No-Asset-Sale (3)	2435.25	2441.31	2425.42

Table 2: **The Bond Values with Three Different Scenarios under the Total-Asset-Sale and No-Asset-Sale Assumptions.** The values of B_2 are calculated by our tree with 1000 time steps. The parameters are from Figs. 6 and 7. (1) and (2) denotes the default is triggered exogenously by the exogenous default boundaries \$2,400 and \$3,000, respectively, whereas (3) denotes the default is triggered endogenously if the firm fails to raise sufficient equity capital to meet current debt obligation.

default is triggered endogenously if the firm fails to raise sufficient equity capital to meet current debt obligation. To our knowledge, only Geske (1977) develops the compound option pricing formula for evaluating the risky bond under the following constraints: (1) no-asset-sale assumption, (2) no positive net-worth covenant, and (3) $B_1 \succ B_2$. The value of B_2 priced by his formula is \$2,425.37, which can be well approximated by the value \$2,425.42 calculated by our tree. Note that the bond values for case (1) and (3) are close because the default boundary \$2,400 is too low in this case. As the default boundary is raised to \$3,000 (i.e., case (2)), the bond values will increase and be different from those in case (1) because of the better protection resulted from the higher exogenous default boundary. As shown in the table, different bond covenants and liability structures of a firm do impact the values of the risky bonds; nevertheless, they cannot be accurately priced by the existing approaches under other bond covenants. In contrast, pricing them by our tree is quite straightforward and simple.

6 Numerical Results

This section examines how the bond covenants, tax benefits, bankruptcy costs, and the jumps of the firm's asset value influence the prices of defaultable bonds. First, we verify the robustness and generality of our tree model by showing that many announced structural models can also be accurately priced by our tree model. We then show how the bond's value is influenced under different asset-sale assumptions and positive net-wroth covenants. Next, we show that how the bond's value is influenced by the repayment of other outstanding bonds of the same issuers. The numerical results suggest that the bond holder can be well-protected from the damage of repaying other outstanding bonds by the restriction on selling the firm's asset value to finance the repayment or by the puttable covenant. Besides, by assuming that the firm's asset value follows the jump-diffusion process, our numerical results suggest that the credit spread of short-term debt are much larger than zero, and the shapes of the term structure of credit spreads can be more varied, which are consistent to empirical studies (see Zhou, 2001).

6.1 Robustness and Generality

Our tree model can accurately evaluate the credit risk for a wide class of structural models by incorporating the extrapolation method for improving the accuracy of numerical methods suggested

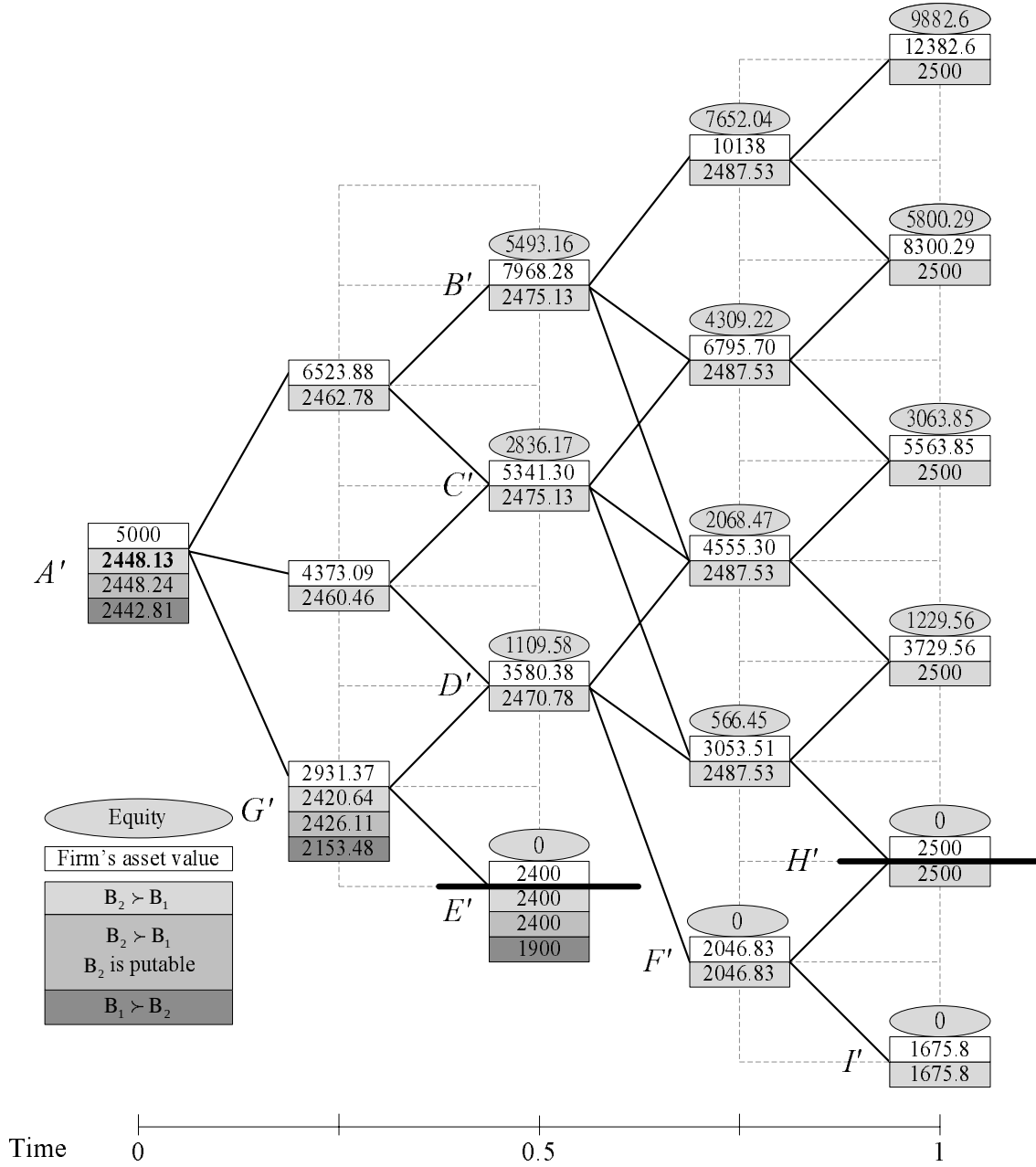


Figure 7: **A Numerical Example for the No-Asset-Sale Assumption.** All the numerical settings are the same as those in Fig. 6 except that the firm is not allowed to sell its assets to finance the repayment of B_1 at year 0.5. The thick black lines denote the default boundaries. The numbers in the ellipses denote the equity values after the loan repayment. The numbers in the white cells denote the firm's asset value. The values of B_2 under Scenario 1 are listed in light-gray cells. The values of B_2 under Scenario 2 and Scenario 3 at nodes A' , E' , and G' are listed in the mid-gray cells and the dark-gray cells, respectively. The values of B_2 under these two scenarios at other nodes are listed in the light-gray cells. The branching probabilities for nodes A' , B' , C' , and D' are listed in Table 1.

in Heston and Zhou (2000). Numerical results illustrated in Table 3 verify the robustness and generality of our tree model by comparing the debt values generated by our tree model and by some important structural models that can be analytically assessed. Merton (1974) assumes that the firm issues a zero-coupon bond and the default event only occurs at bond's maturity when its asset value cannot meet the bond repayment. Thus the bond price can be evaluated by the wellknown Black-Scholes formula (see Black and Scholes, 1973). Here we assume that the firm issues a zero-coupon bond with one year maturity and face value \$3,000. The numerical results listed in the "Merton (1974)" column verify that our tree model can generate accurate bond prices under different scenarios. Black and Cox (1976) extends Merton's model by assuming that the firm can default prior to its maturity if its asset value is less than the exogenous default boundary set as the discounted present value of debt's face value. Here we assume the discount rate of the exogenous default boundary is 4%. In "Black and Cox (1976)" column, the values generated by our tree again accurately approximate the values priced by the Black and Cox (1976) formula.

Leland (1994) analyzes the trade-off theory (Kraus and Litzenberger, 1973) and optimal capital structure by considering the effects of the tax benefit and bankruptcy cost. In "Leland (1994)" column, the firm is assumed to issue a consol bond with optimal coupon payments. The default is triggered endogenously if the firm fails to raise sufficient equity capital to meet current debt obligation. Although our tree model can only evaluate a bond with finite maturity, our tree can accurately evaluate the maximum debt value (or the debt capacity in Leland (1994)) by setting the finite maturity of the bond T to a large number, says $T = 200$. Similar phenomenon is also mentioned in Brennan and Schwartz (1978).

Geske (1977) investigates the effects of complex liability structures under a very restricted case: the firm issues one short-term senior zero-coupon bond and one long-term junior zero-coupon bond. Under the no-asset-sale assumption, he derives a compound option pricing formula to evaluate the long-term junior bond. In "Geske (1977)" column, we assume that a bond B_1 with face value \$500 matures at year 0.5, another bond B_2 with $F_2 = \$2,500$ matures at year 1, and B_1 is more senior than B_2 (*i.e.*, $B_1 \succ B_2$). Note that the default event occurs when the firm fails to raise sufficient equity capital to meet the repayment of B_1 at year 0.5 or when the firm's value fails to meet the debt obligation B_2 at year 1. The bond prices generated by our tree well approximate the prices generated by Geske's formula. In addition, the following numerical examples will show that our tree model can be easily extended to other general cases with complex liability structures and bond covenants, which can not be efficiently or accurately evaluated by existing formulas and numerical methods.

6.2 Asset Sale Assumptions and Existence of Positive Net-worth Bond Covenant

Some bond covenants protect the bond holders by restricting selling firm's asset to finance the bond repayments or by enforcing an exogenous default boundary. We compare the effects of these protection covenants under different bankruptcy cost α and tax benefit τ in Table 4. We follow Leland (1994) by naming a debt without positive net-worth covenant as a unprotected debt. The default for a unprotected debt is triggered endogenously if the firm fails to raise sufficient equity capital to meet current debt obligation. On the other hand, for the protected debt, the firm defaults once its asset value falls below the exogenous default boundary. In this case, we follow Brennan and Schwartz (1978) by assuming that the default boundary is a constant fraction, says 90% of the face

σ_V	Merton (1974)		Black and Cox (1976)		Leland (1994)		Geske (1977)	
	Tree	Formula	Tree	Formula	Tree	Formula	Tree	Formula
0.25	2934.82	2934.82	2940.03	2940.03	3419.57	3419.38	2449.81	2449.79
	(0.00003%)		(0.00002%)		(−0.00556%)		(−0.00082%)	
0.4	2875.60	2875.60	2935.53	2935.53	2941.41	2942.23	2425.37	2425.55
	(0.00014%)		(−0.00010%)		(−0.02788%)		(−0.00742%)	

Table 3: Robustness of Our Tree Model. The initial firm value V is \$5,000, the risk-free interest rate r is $= 2\%$, and the jump intensity λ is 0. The first column σ_V denotes the volatility of the firm’s value. The values under “Tree” and “Formula” denote the bond values generated by our tree model and by analytical formulas. The values in the parentheses denote the relative pricing errors. In “Merton (1974)” and “Black and Cox (1976)” columns, the firm is assumed to issue a zero-coupon bond with one year maturity and face value \$3,000, both the bankruptcy cost α , and the tax rate τ are assumed to be 0. The exogenous default boundary under Black and Cox (1976) model at time t is set to $3000e^{-\gamma(1-t)}$, where the discount rate γ is 4%. In “Leland (1994)” column, the firm is assumed to issue a consol bond with the optimal coupon per year $C = \$112.259$ when $\sigma_V = 0.25$ and $C = \$112.259$ when $\sigma_V = 0.4$. The tax rate τ and the bankruptcy cost α are assumed to be 0.35 and 0.5, respectively. Note that the default is triggered endogenously if the firm fails to raise sufficient equity capital to meet current debt obligation, which is the after-tax coupon $(1 - \tau)C$. In “Geske (1977)” column, the firm is assumed to issue two zero-coupon bonds: B_1 with face value \$500 matures at year 0.5 and B_2 with $F_2 = \$2,500$ matures at year 1 ($B_1 \succ B_2$). Both the bankruptcy cost α and the tax rate τ are set to 0. The default event occurs when the firm fails to raise sufficient equity capital to meet the debt repayment \$500 at year 0.5 or when the firm’s value fails to meet the debt obligation \$2,500 at year 1. Note that we list the values of bond B_2 in this column.

value (i.e, the default boundary is $0.9 \times \$3,000 = 2,700$).

It can be observed that the debt value increases with the tax benefit and decreases with the bankruptcy cost. Furthermore, when no bankruptcy cost is presented (i.e, $\alpha = 0$), the protected debt values are always higher than the unprotected ones. However, the situation can be reversed with high bankruptcy cost, says $\alpha = 0.5$. This is because the existence of exogenous default boundary increases the default probability, and part of firm’s asset ($\alpha\%$ in this example) flows to liquidator or lawyer instead of equity shareholders or bond holders.

This table also shows how asset-sale assumptions influence debt values. The impact can be very significant; For example, the difference between the values of the protected debt under the no-asset-sale and total-asset-sale assumptions can be as large as \$69 ($\approx 2955.14 - 2886.09$) when $\alpha = 0.5$ and $\tau = 0$. In addition, the bond values under the no-asset-sale assumption for both unprotected and protected debts are higher than those under the proportional-asset-sale and total-asset-sale assumptions. The reason is that under the no-asset-sale assumption, bond holders have better protection than the protection under the other two asset-sale assumptions. Note that pricing a risky bond under the total-asset-sale assumption with complex bond covenants is difficult both analytically or numerically. Fortunately, our tree model can address this problem without difficulty.

6.3 Complex Liability Structures

We now discuss the values of debts with complex liability structures. Assume a firm issues two zero-coupon bonds: B_1 with face value \$500 matures at year 2.99, 3, and 3.01 and B_2 with $F_2 = \$2,500$ matures at year 3.

α	τ	Unprotected debt			Protected debt		
		Total	Proportional	No	Total	Proportional	No
0	0	3135.68	3143.18	3159.66	3141.11	3144.91	3159.95
0.5	0	2904.07	2916.60	2961.33	2886.09	2904.68	2955.14
0	0.35	3145.49	3149.52	3159.85	3147.05	3150.42	3159.96
0.5	0.35	2937.22	2945.26	2974.82	2921.15	2930.30	2961.82

Table 4: **Comparisons on Different Assumptions on Asset Sales.** The initial firm value is $V = \$5,000$, the risk-free interest rate is $r = 2\%$, the firm volatility is $\sigma_V = 40\%$, and the jump intensity is $\lambda = 0$. The bankruptcy cost α and the tax benefit τ are listed in the first and in the second columns, respectively. The firm is assumed to issue a coupon bond with one year maturity, face value $\$3,000$, and annualized coupon rate 10% . The firm is assumed to be monitored quarterly and the coupon is also assumed to be paid quarterly. The value of the risk-free bond with face value $\$3,000$ and annualized coupon rate 10% is $\$3,237.62$.

We first consider the case that the debts are unprotected and the default is triggered endogenously if the firm fails to raise sufficient equity capital to meet current debt obligation under the no-asset-sale assumption. Table 5 lists the credit spreads of B_1 priced by two existing approaches and our tree model. The formulas proposed by Geske (1977) can evaluate the bonds under the assumption that the bond with shorter maturity is more senior than that with longer one. Additionally, when the maturities of the two bonds are the same, the bonds can be priced by the Black-Scholes formula (see Black and Scholes, 1973; Lando, 2004). Clearly, the credit spreads priced by the formulas are all close to those calculated by our tree. Nevertheless, when the bond with longer maturity is more senior than that with shorter one, the bonds cannot be priced by both existing formulas; however, they can be evaluated by our tree model.

We then consider the case that the debts are protected and the default is triggered by an exogenous default boundary. Table 6 tabulates the credit spreads of B_1 priced by our tree under the total-asset-sale and no-asset-sale assumptions. First, we analyze the case that B_1 is more senior than B_2 . Under the total-asset-sale assumption, even if $B_1 \succ B_2$, the credit spread of the non-puttable bond B_1 at year 3.01 jumps from 1.89×10^{-3} (bps) to 40.86 (bps) because repaying B_2 at year 3 by selling the firm's asset significantly increases the risk of B_1 . The results also show that the risk can be greatly reduced by adding the puttable feature in B_1 ; for example, the credit spreads for the puttable bond at year 3.01 is 2.89×10^{-2} , which is much lower than 40.86 for the non-puttable one. Under the no-asset-sale assumption, the credit spread at year 3.01 decreases from 1.89×10^{-3} (bps) to 7×10^{-4} (bps) since this assumption also provides protection for the bond holders. In this case, the puttable feature does not have significant impacts on the bond values. On the other hand, when B_2 is more senior than B_1 , the risk of B_1 becomes much higher than that under the case that $B_1 \succ B_2$.

Under the total-asset-sale assumption, bonds can be evaluated by the approximate analytical formulas for pricing stock options with dividend payout. Recall that the firm's asset value and the payout by selling the firm's asset are mapped to the stock price and dividend payout, respectively. There are two ways to deal with the payout: (1) subtracting the present value of the face value of the debt with shorter maturity (the payout) from the initial firm's asset value; (2) treating the payout as continuous dividends.¹⁰ Note that the default boundary will also be adjusted to proportion of the face value of the debt with longer maturity. However, these approximations lead to unreasonable

¹⁰The continuous dividend yield is calculated by the method mentioned in the footnote of Hernández et al. (2007).

Maturity	The values of bond B_1			
	$B_1 \succ B_2$		$B_2 \succ B_1$	
	Formula	Tree	Formula	Tree
2.99	470.81	470.81	x	342.38
3	470.80	470.80	342.27	342.27
3.01	x	470.70	342.20	342.22

Table 5: **The Values of Bond B_1 Given Bond B_2 with a Fixed Maturity When There is no Positive Net-Worth Covenant.** The initial firm value is $V = \$5,000$, the risk-free interest rate is $r = 2\%$, the firm volatility is $\sigma_V = 40\%$, the jump intensity is $\lambda = 0$, the bankruptcy cost is $\alpha = 0$, and the tax rate is $\tau = 0$. The firm is assumed to issue two zero-coupon bonds: B_1 with face value \$500 matures at year 2.99, 3, 3.01 and B_2 with $F_2 = \$2,500$ matures at year 3. The firm value is monitored at payment due dates and the default is assumed to be triggered endogenously if the firm fails to raise sufficient equity capital to meet current debt obligation. When the maturities of the two bonds are different and the bond with shorter maturity is more senior than that with longer maturity, they are priced by the formulas proposed in Geske (1977), whereas when those of the two bonds are the same, the bonds are priced by the Black-Scholes formula (see Black and Scholes, 1973; Lando, 2004).

bond prices because they treat the bond with shorter maturity as a risk-free bond. For example, the credit spreads for the bonds maturing at year 3.01 generated by methods (1) and (2) are 9.406 and 10.498, respectively, which are both much lower than 1613.553. The above results suggest that our tree model not only accurately evaluates the bonds with complex liability structures under different assumptions on asset sales but also gives more realistic bond prices than the existing approximate method.

6.4 Stochastic Jump in Firm Value

It is known that a lognormal diffusion process has a continuous sample path and cannot cross a default boundary from somewhere else instantaneously. Therefore, under a diffusion process, if a firm is not currently in financial distress, the credit spreads of very short-term debts are nearly zero. But in the real world, the credit spreads of short-term debts are often larger than zero. If the evolution of the firm value follows a jump-diffusion process, however, the story will be different. Under a jump-diffusion process, a default can happen instantaneously because they may have a sudden jump in firm's asset value. As a result, the credit spreads of short-term debts generated by a jump-diffusion model will be much larger than zero.

This paper also incorporates the jump-diffusion process into our tree model. Figure 8(a) compares the credit spreads under the lognormal diffusion process and the jump-diffusion process. As shown in Fig. 8, the credit spreads for the short-term debts under the jump-diffusion process are indeed larger than those under the lognormal diffusion process, which are nearly zero. Figure 8(b) displays the effect of the source of volatility on credit spreads. The figure illustrates that the jump-diffusion process can generate rich dynamics of the term structure of credit spreads, such as humped, flat, or reversed curves.

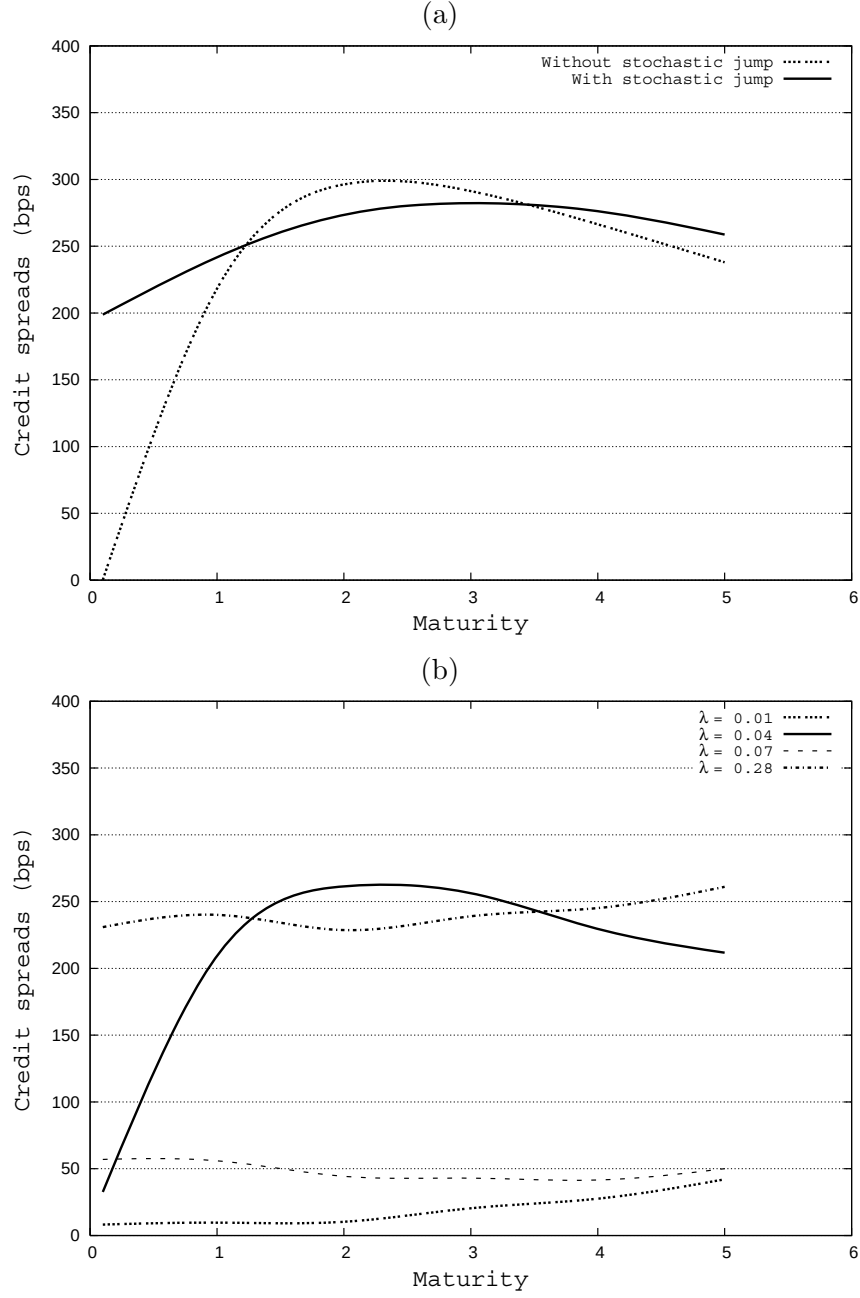


Figure 8: The Effect of the Stochastic Jump. The initial firm value is $V = \$5,000$, the risk-free interest rate is $r = 2\%$, the bankruptcy cost is $\alpha = 0$, and the tax rate is $\tau = 0$. The firm is assumed to issue a zero-bond with one year maturity and face value $\$3,000$; additionally, the exogenous default boundary is set to $\$2,400$ ($0.8 \times \$3,000$). The firm value is monitored quarterly. (a) The parameters for the model without the stochastic jump are $\sigma_V = 40\%$, $\lambda = 0$ and $\delta^2 = 0$, and those for the model with the stochastic jump are $\sigma_V = 40\%$, $\lambda = 1$ and $\delta^2 = 0.15$. (b) The parameters are $\delta^2 = 0.5$, and $\sigma^2 = \sigma_V^2 - \lambda\delta^2$ (λ is adjusted by this formula); the dotted and dashed curves denote the credit spreads under firm volatility $\sigma_V^2 = 20\%$, and the solid and dash-dotted ones denote those under $\sigma_V = 40\%$.

The credit spreads (bps) of bond B ₁				
Total-Asset-Sale				
Maturity	B ₁ $\succ$ B ₂		B ₂ $\succ$ B ₁	
	Non-putable	Putable	Non-putable	Putable
2.99	0.00002	0.00002	1553.54333	42.35894
3	0.00189	0.00189	1622.31542	42.51885
3.01	40.86288	0.02893	1613.55337	42.67799
No-Asset-Sale				
Maturity	B ₁ $\succ$ B ₂		B ₂ $\succ$ B ₁	
	Non-putable	Putable	Non-putable	Putable
2.99	0.00002	0.00002	1612.11056	42.38770
3	0.00189	0.00189	1622.31542	42.53319
3.01	0.00070	0.00070	1611.75318	42.69228

Table 6: **The Credit Spreads of Bond B₁ Given Bond B₂ with a Fixed Maturity When There is a Positive Net-Worth Covenant.** The initial firm value is $V = \$5,000$, the risk-free interest rate is $r = 2\%$, the firm volatility is $\sigma_V = 40\%$, the jump intensity is $\lambda = 0$, the bankruptcy cost is $\alpha = 0$, and the tax rate is $\tau = 0$. The firm is assumed to issue two zero-coupon bonds: B₁ with face value \$500 matures at year 2.99, 3, 3.01 and B₂ with $F_2 = \$2,500$ matures at year 3. The put price for B₁ is $500e^{-\kappa(T-t)}$, where $\kappa = 4\%$. The firm value is continuously monitored and the default is assumed to be triggered by the exogenous default boundary \$2,400.

7 Conclusions

This paper prices risky bonds by incorporating the complex liability structure and bond covenants into the structural model. An efficient tree model is then presented to price risky bonds. Because of its accuracy, this numerical model is of great help to exploring how credit spreads are influenced by the debt covenants and the change of the liability structure. Furthermore, our tree can be extended to model the jump-diffusion process, which can fit the empirical data better for its ability to generate large credit spreads for short-term corporate bonds and rich dynamics of the term structure of credit spreads. The numerical results not only confirm the robustness and generality of our tree model but also show its ability to accurately evaluate the risky bonds with complex liability structures and bond covenants.

A Validity of Risk-Neutral Probabilities

Define

$$\begin{aligned}
\det &= (\beta - \alpha)(\gamma - \alpha)(\gamma - \beta), \\
\det_u &= (\beta\gamma + \text{Var})(\gamma - \beta), \\
\det_m &= (\alpha\gamma + \text{Var})(\alpha - \gamma), \\
\det_d &= (\alpha\beta + \text{Var})(\beta - \alpha).
\end{aligned}$$

Cramer's rule applied to Eqs. (9) to (11) gives $p_u = \det_u/\det$, $p_m = \det_m/\det$, and $p_d = \det_d/\det$. Note that $\det < 0$ because $\alpha > \beta > \gamma$. To ensure that the branching probabilities are valid, it suffices to show that $p_u, p_m, p_d \geq 0$. As $\det < 0$, it is sufficient to show $\det_u, \det_m$, and $\det_d \leq 0$ instead.

Finally, as $\alpha > \beta > \gamma$, it suffices to show that $\beta\gamma + \text{Var} \geq 0$, $\alpha\gamma + \text{Var} \leq 0$, and $\alpha\beta + \text{Var} \geq 0$ under the premise $\beta \in [-\sigma\sqrt{\Delta t}, \sigma\sqrt{\Delta t})$. Indeed,

$$\begin{aligned}\beta\gamma + \text{Var} &= \beta^2 - 2\beta\sigma\sqrt{\Delta t} + \sigma^2\Delta t' \geq \beta^2 - 2\beta\sigma\sqrt{\Delta t} + \sigma^2\Delta t = (\beta - \sigma\sqrt{\Delta t})^2 \geq 0, \\ \alpha\gamma + \text{Var} &= \beta^2 - 4\sigma^2\Delta t + \sigma^2\Delta t' \leq \beta^2 - 4\sigma^2\Delta t + 2\sigma^2\Delta t = \beta^2 - 2\sigma^2\Delta t \leq 0, \\ \alpha\beta + \text{Var} &= \beta^2 + 2\beta\sigma\sqrt{\Delta t} + \sigma^2\Delta t' \geq \beta^2 + 2\beta\sigma\sqrt{\Delta t} + \sigma^2\Delta t = (\beta + \sigma\sqrt{\Delta t})^2 \geq 0,\end{aligned}$$

as desired.

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